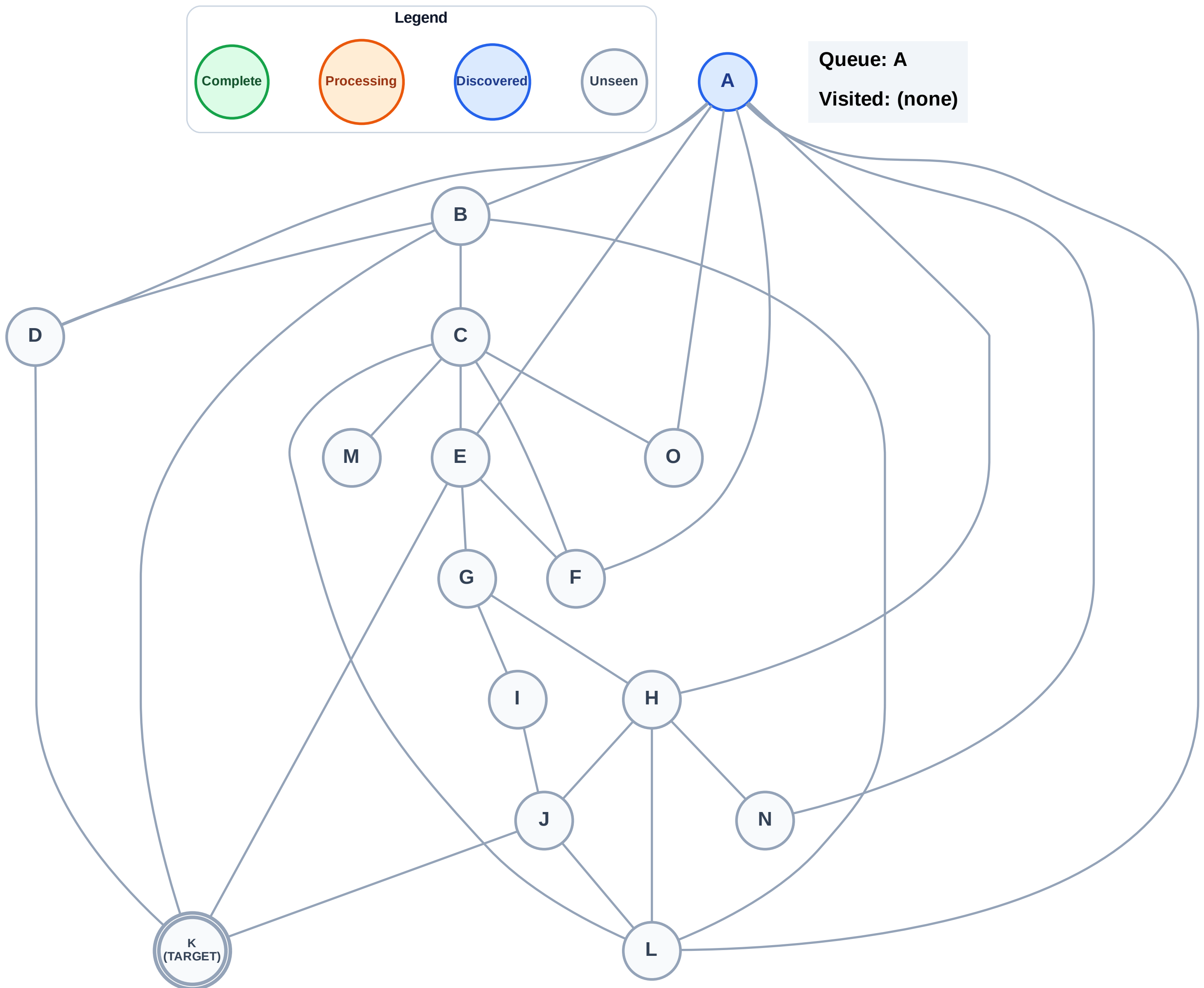


Step 1: Start at A (Seeking target node K)

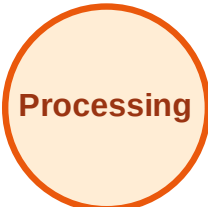
Step 1: Start at A (Seeking target node K)



BFS Traversal

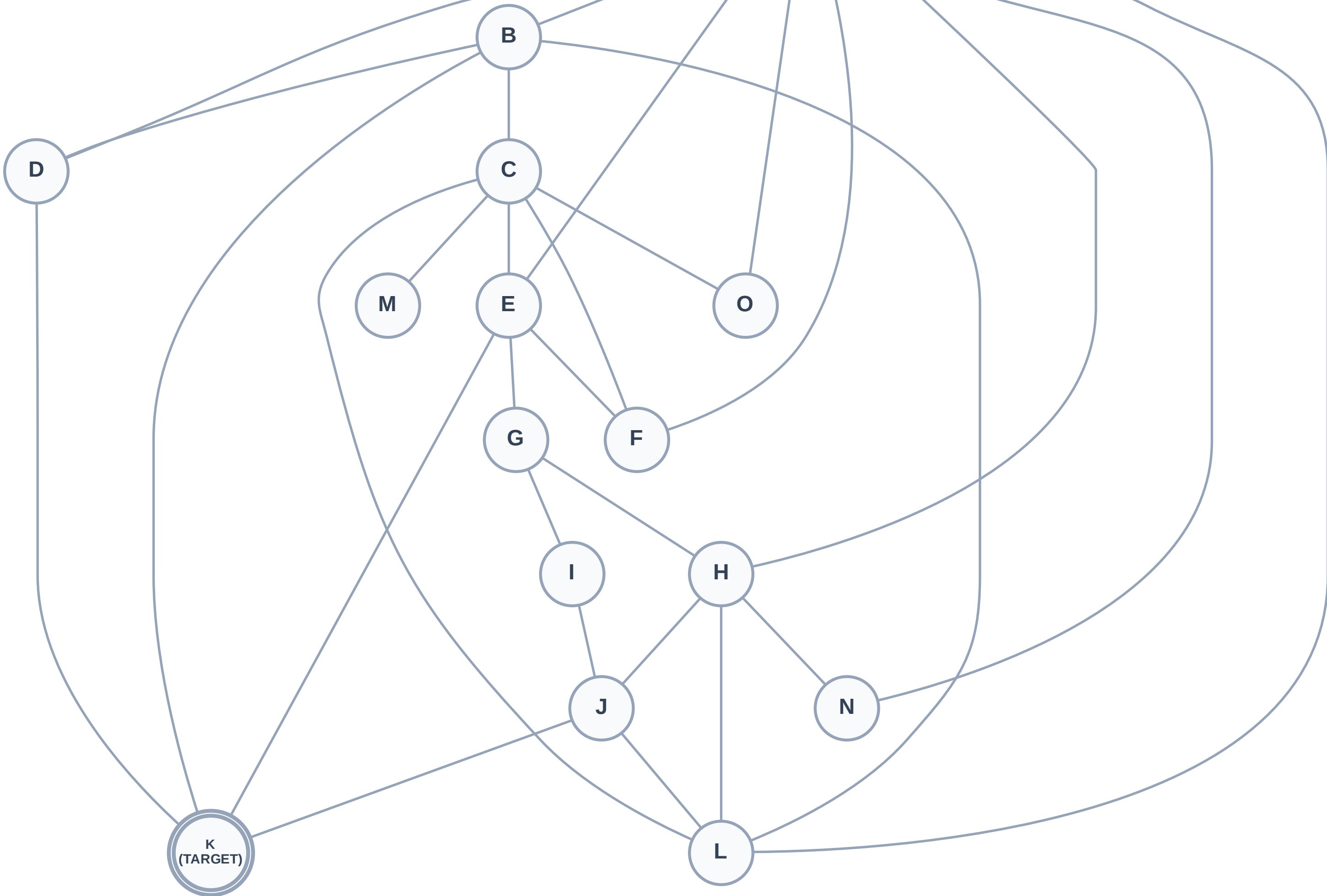
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, E, F, H, L, N, O from A

Legend

Complete

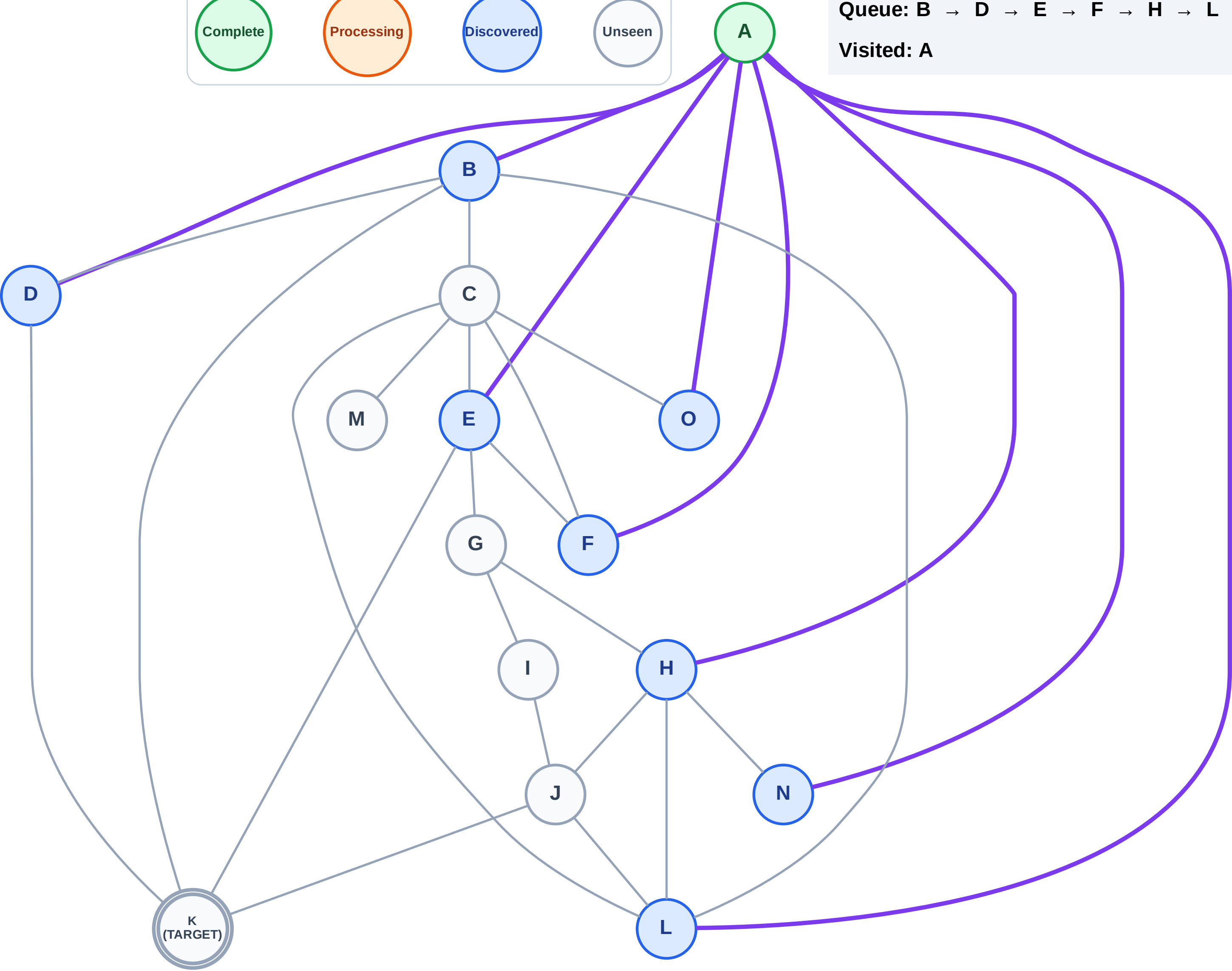
Processing

Discovered

Unseen

Queue: B → D → E → F → H → L → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

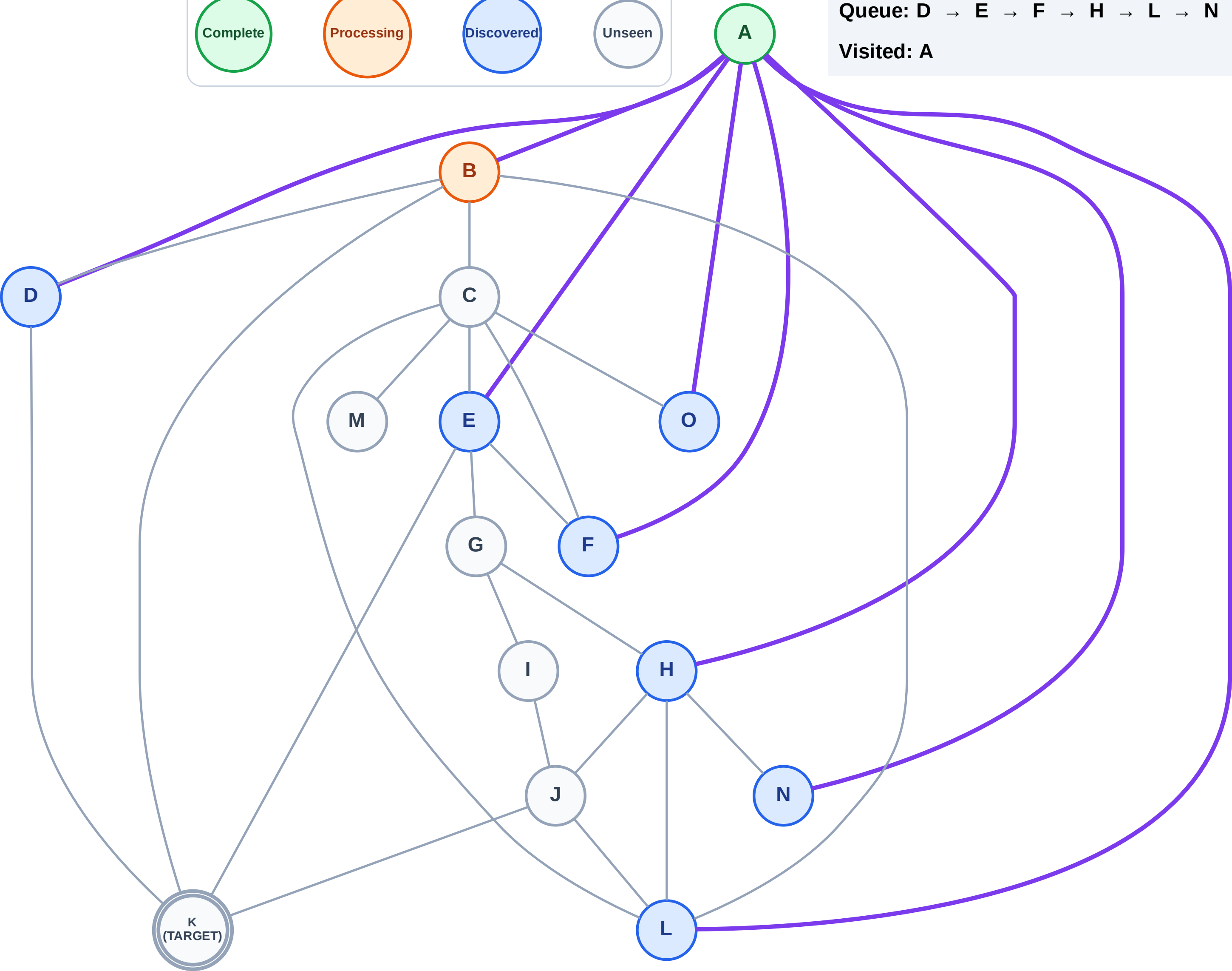
Processing

Discovered

Unseen

Queue: D → E → F → H → L → N → O

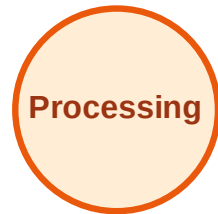
Visited: A



BFS Traversal

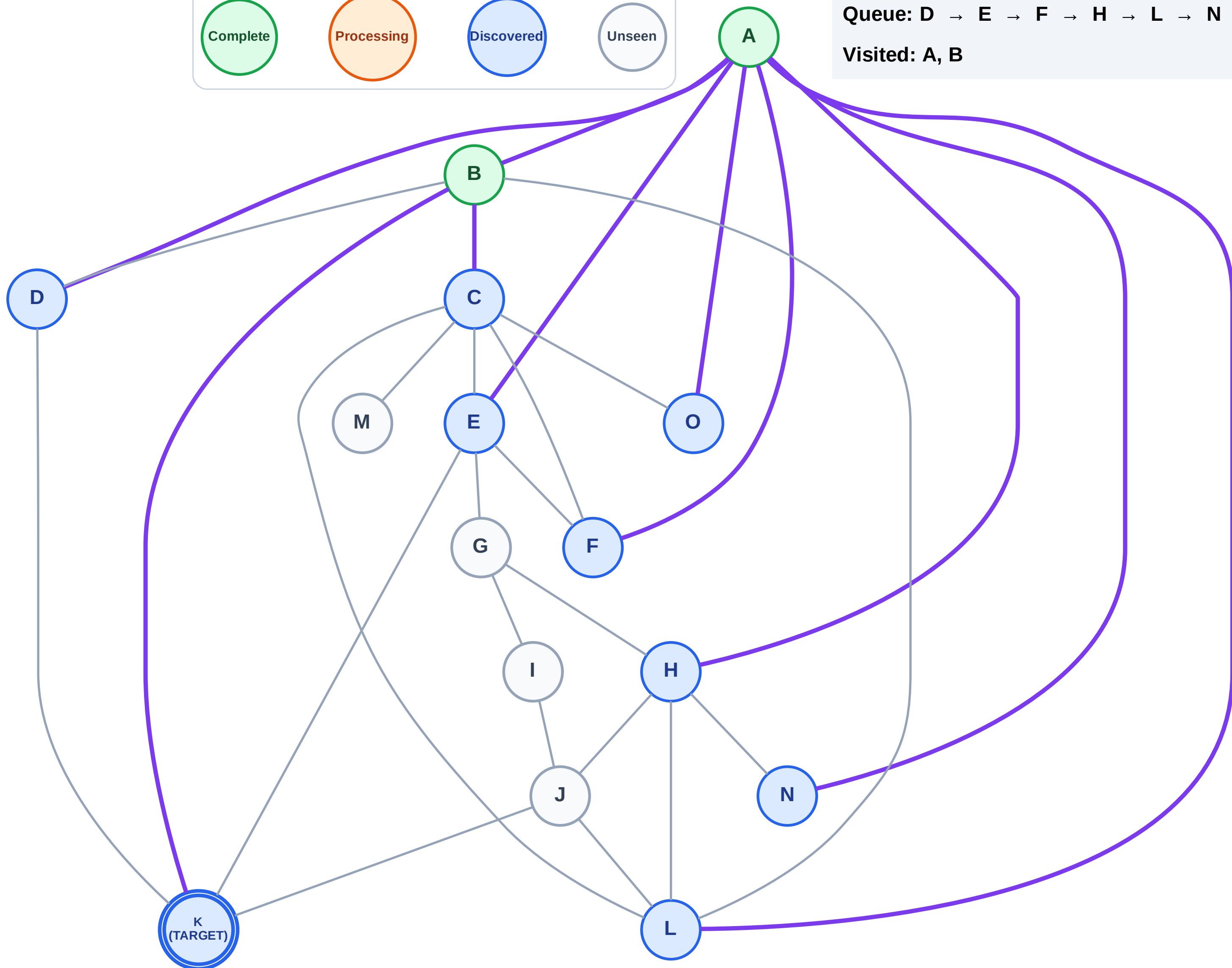
Step 5: Discover C, K from B

Legend



Queue: D → E → F → H → L → N → O → C → K

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

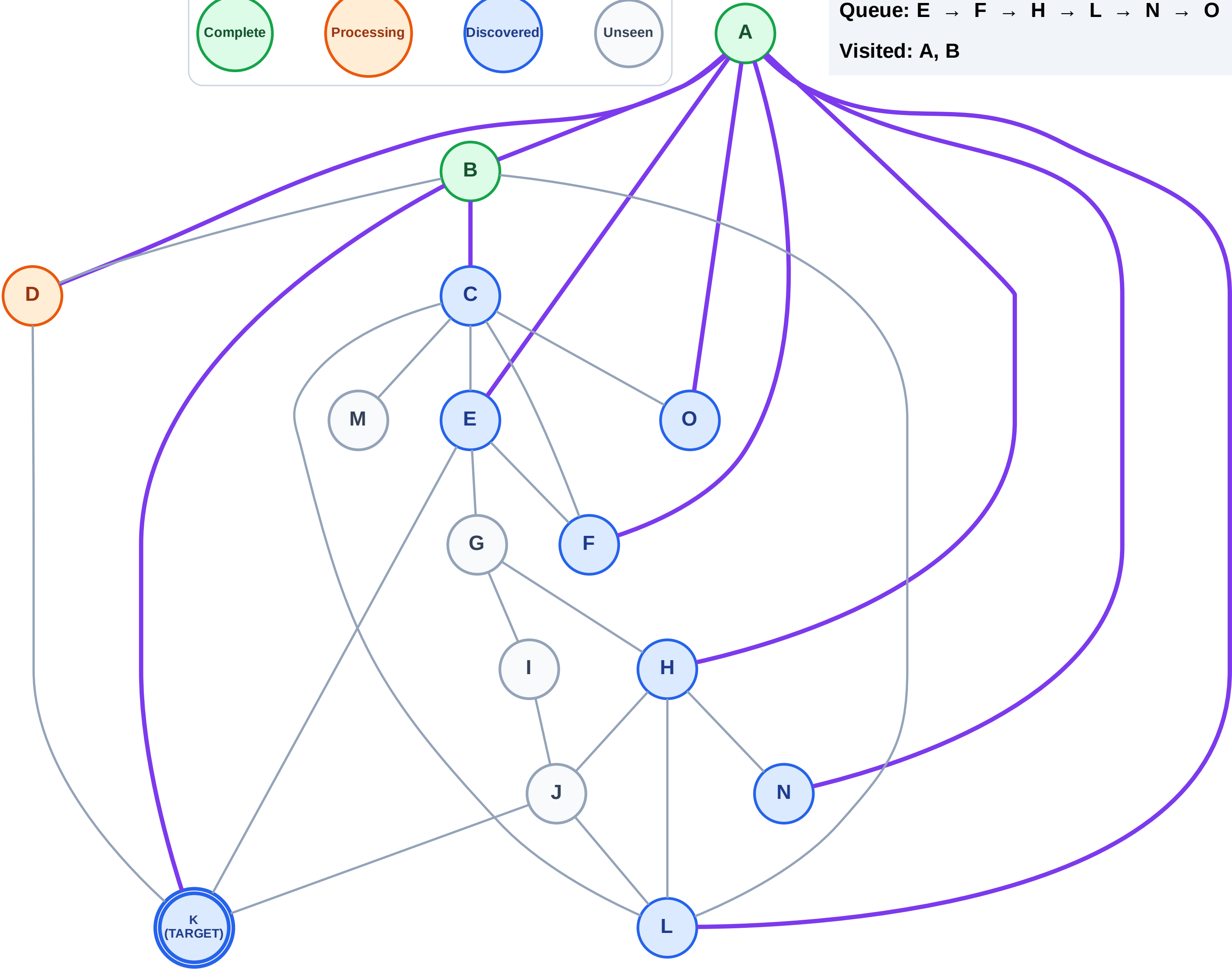
Processing

Discovered

Unseen

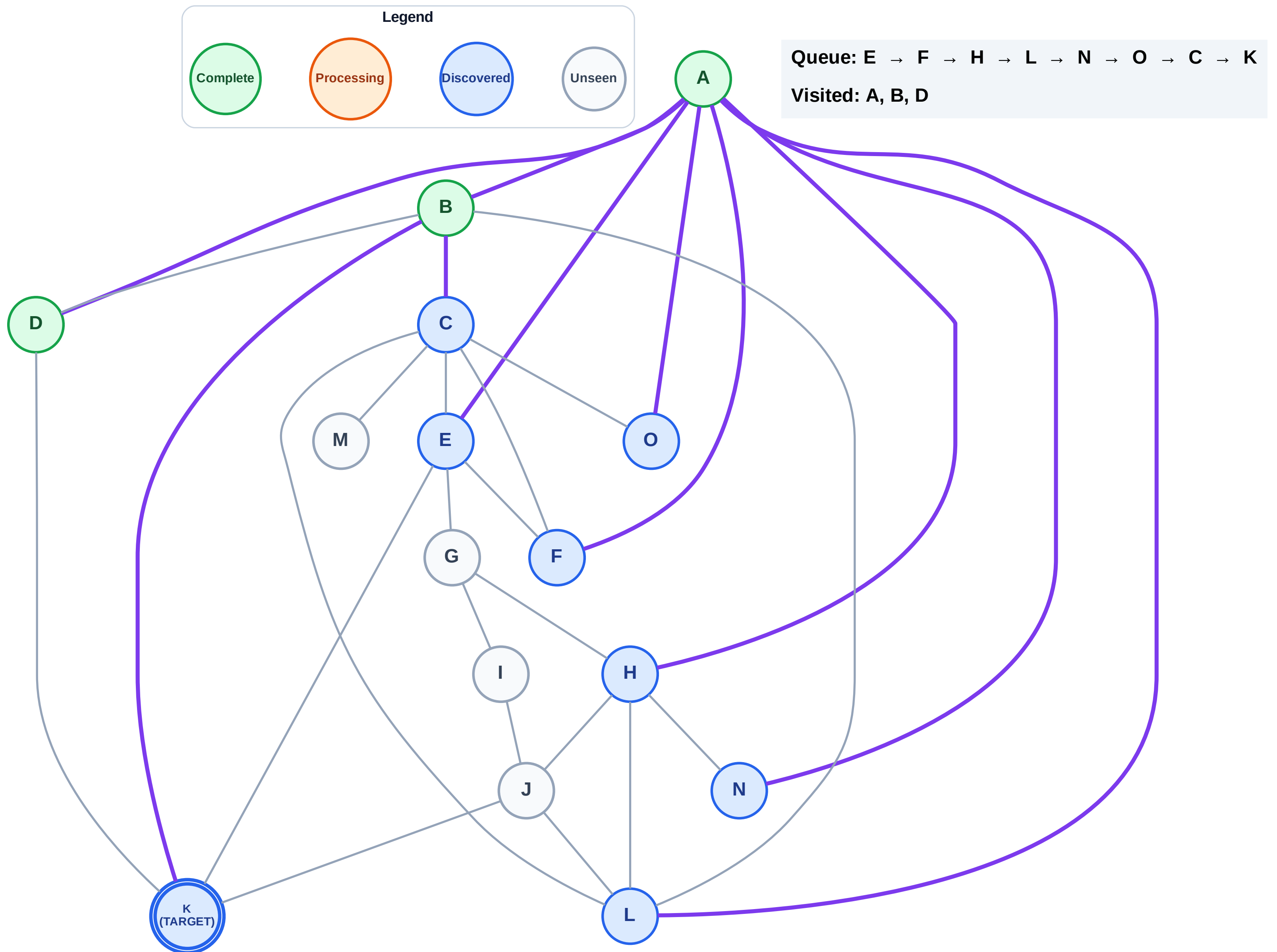
Queue: E → F → H → L → N → O → C → K

Visited: A, B



Step 7: No new neighbors from D

Step 7: No new neighbors from D



BFS Traversal

Step 8: Process node E

Legend

Complete

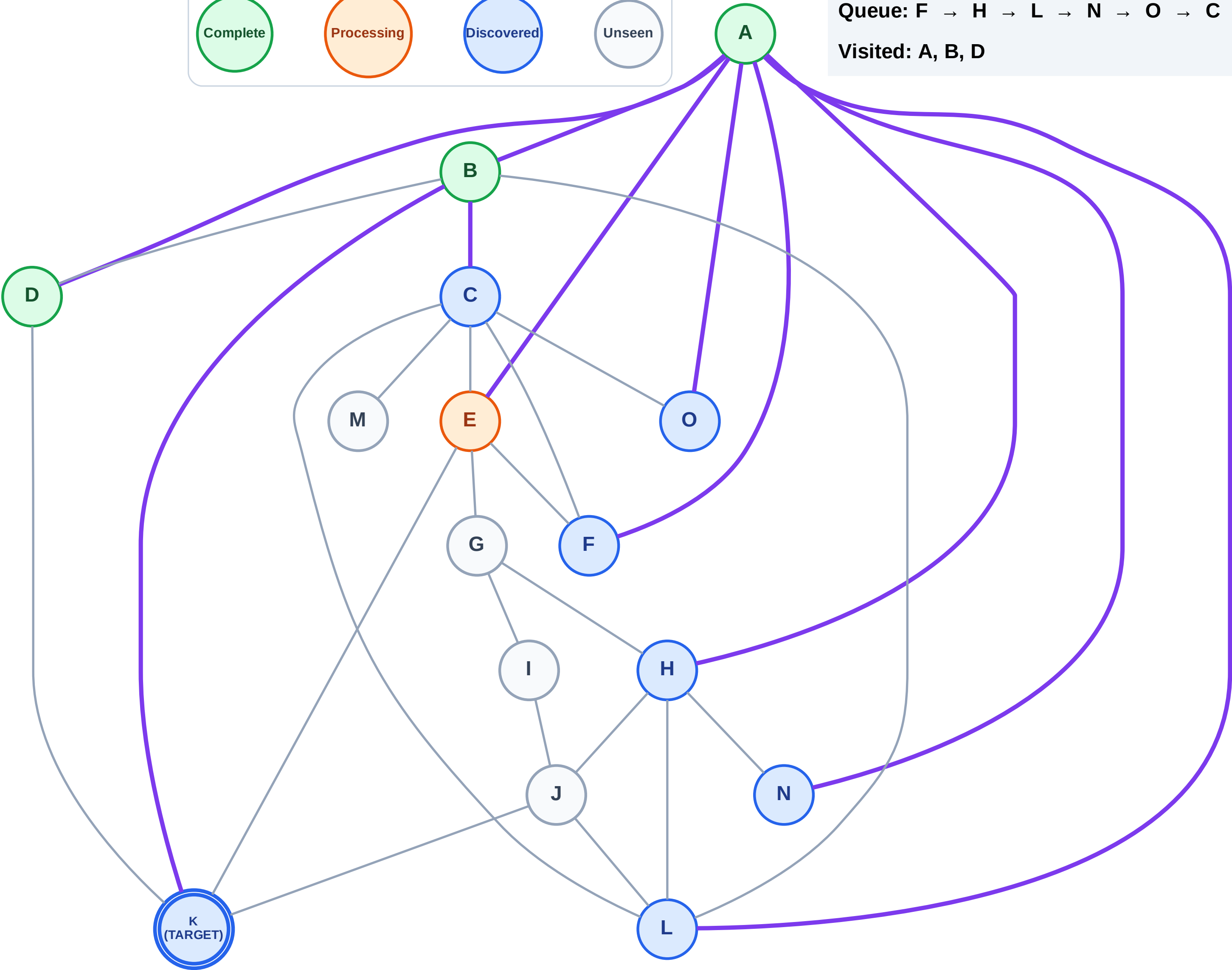
Processing

Discovered

Unseen

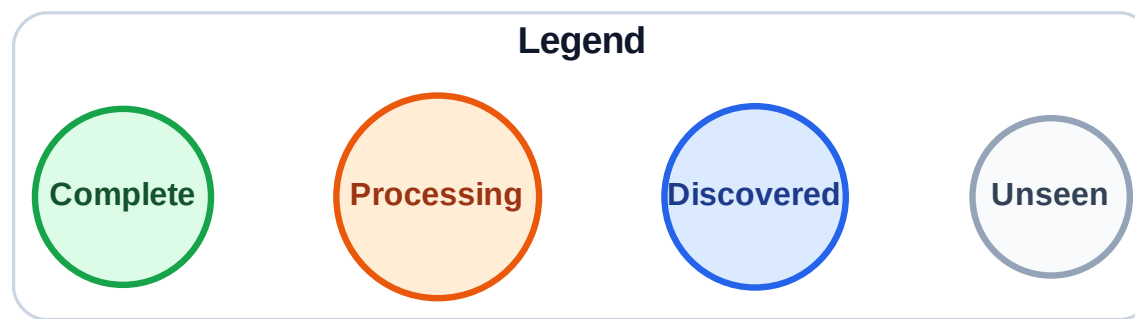
Queue: F → H → L → N → O → C → K

Visited: A, B, D

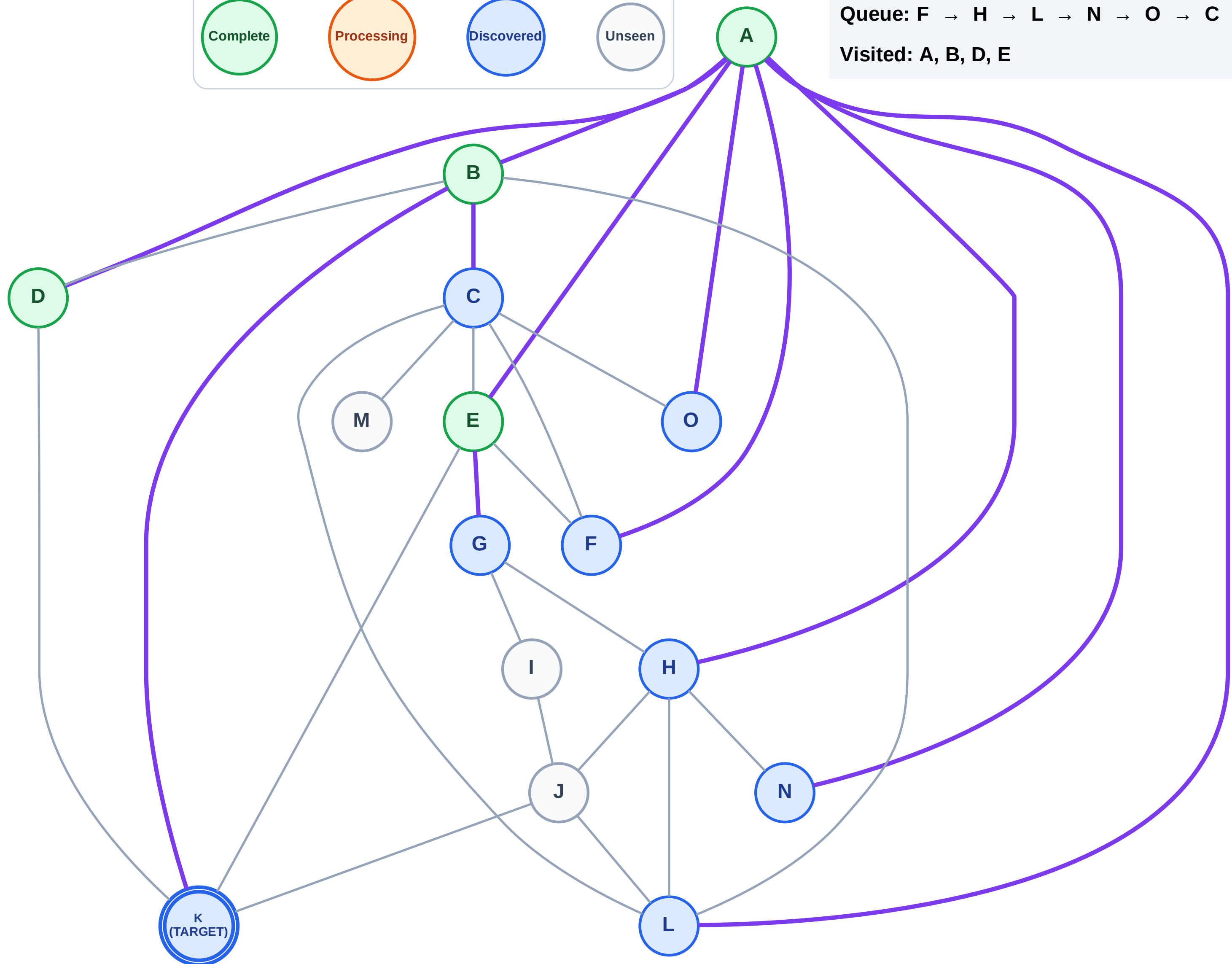


BFS Traversal

Step 9: Discover G from E



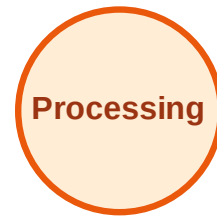
Queue: F → H → L → N → O → C → K → G
Visited: A, B, D, E



BFS Traversal

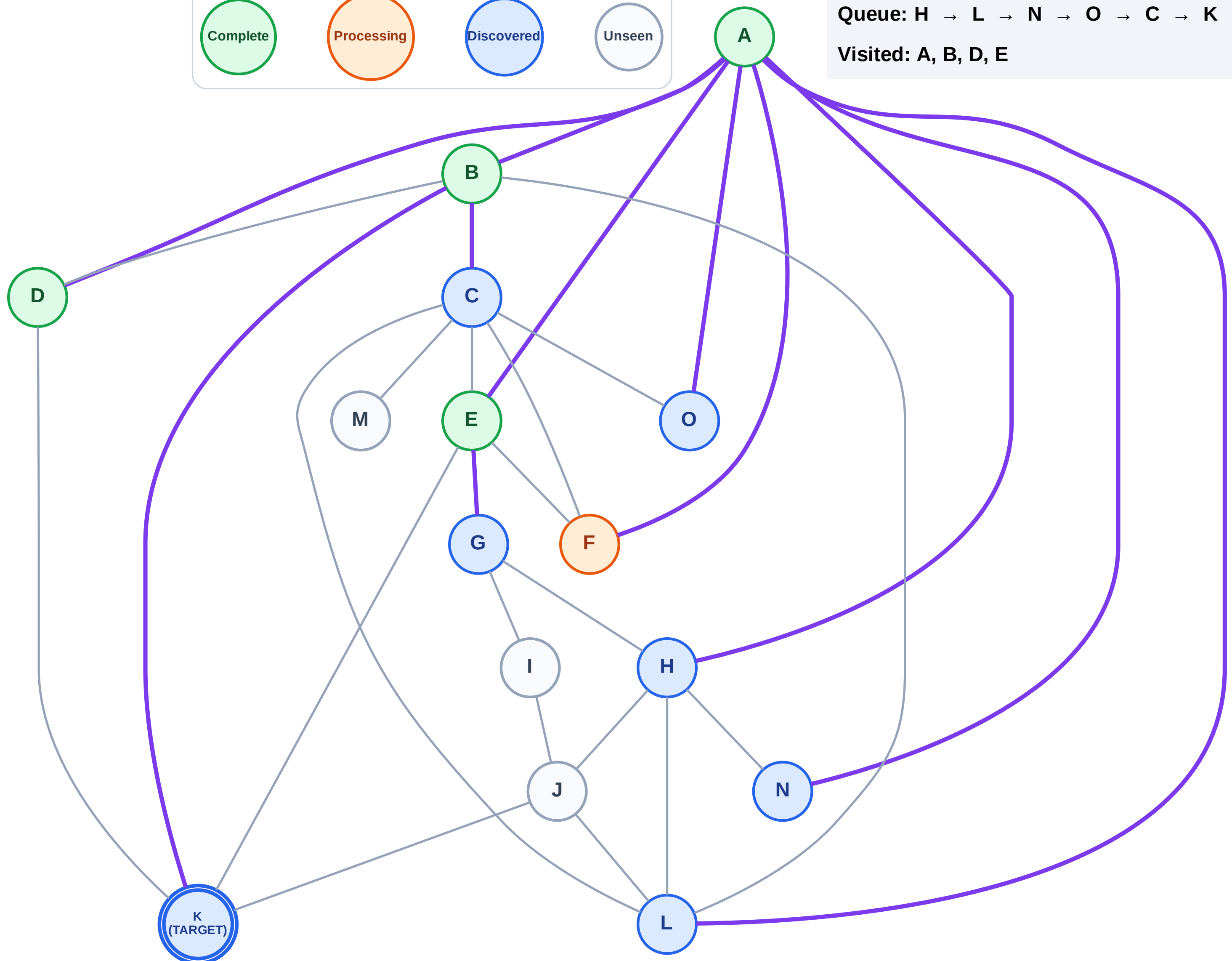
Step 10: Process node F

Legend



Queue: H → L → N → O → C → K → G

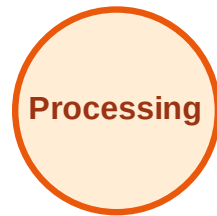
Visited: A, B, D, E



BFS Traversal

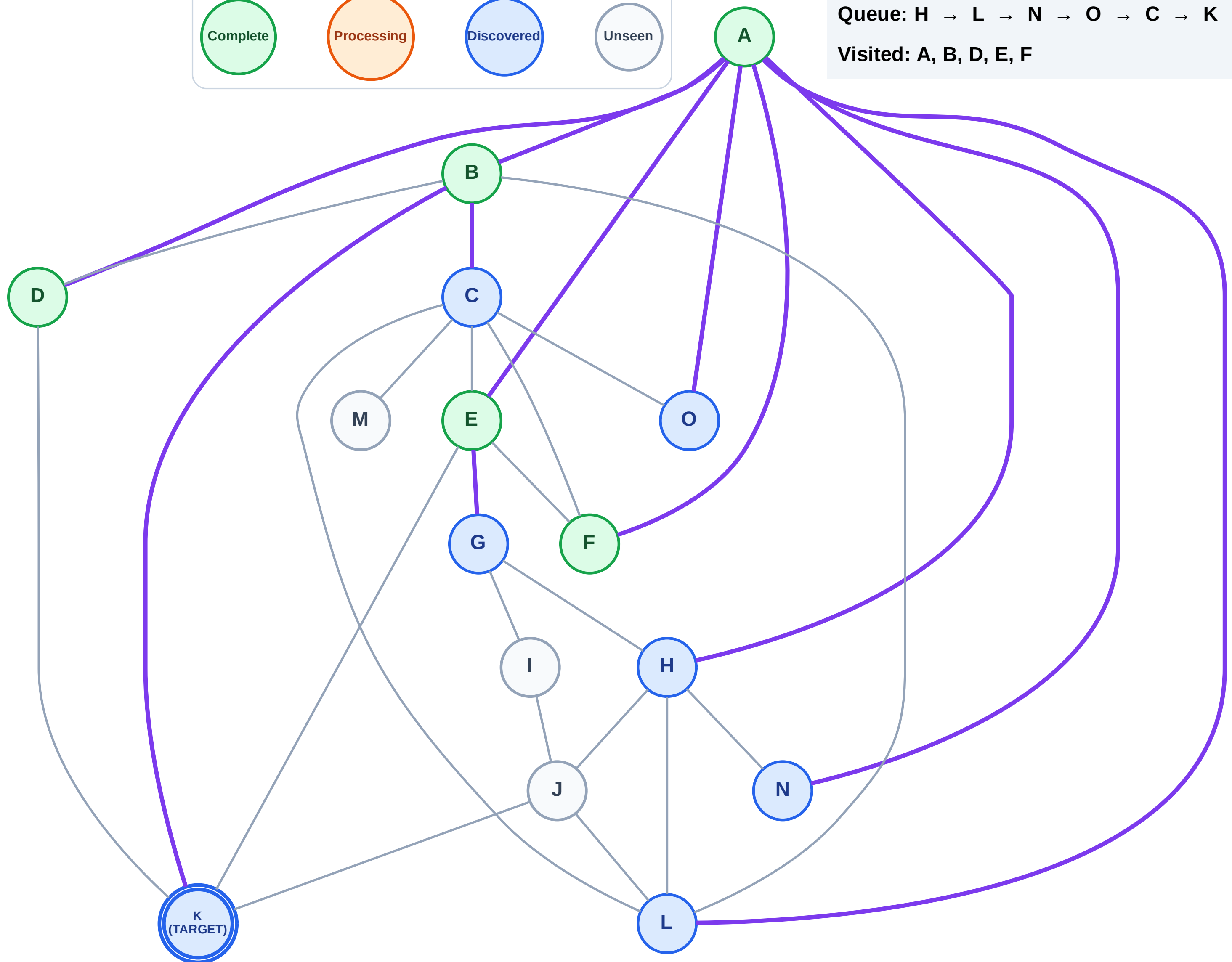
Step 11: No new neighbors from F

Legend



Queue: H → L → N → O → C → K → G

Visited: A, B, D, E, F



BFS Traversal

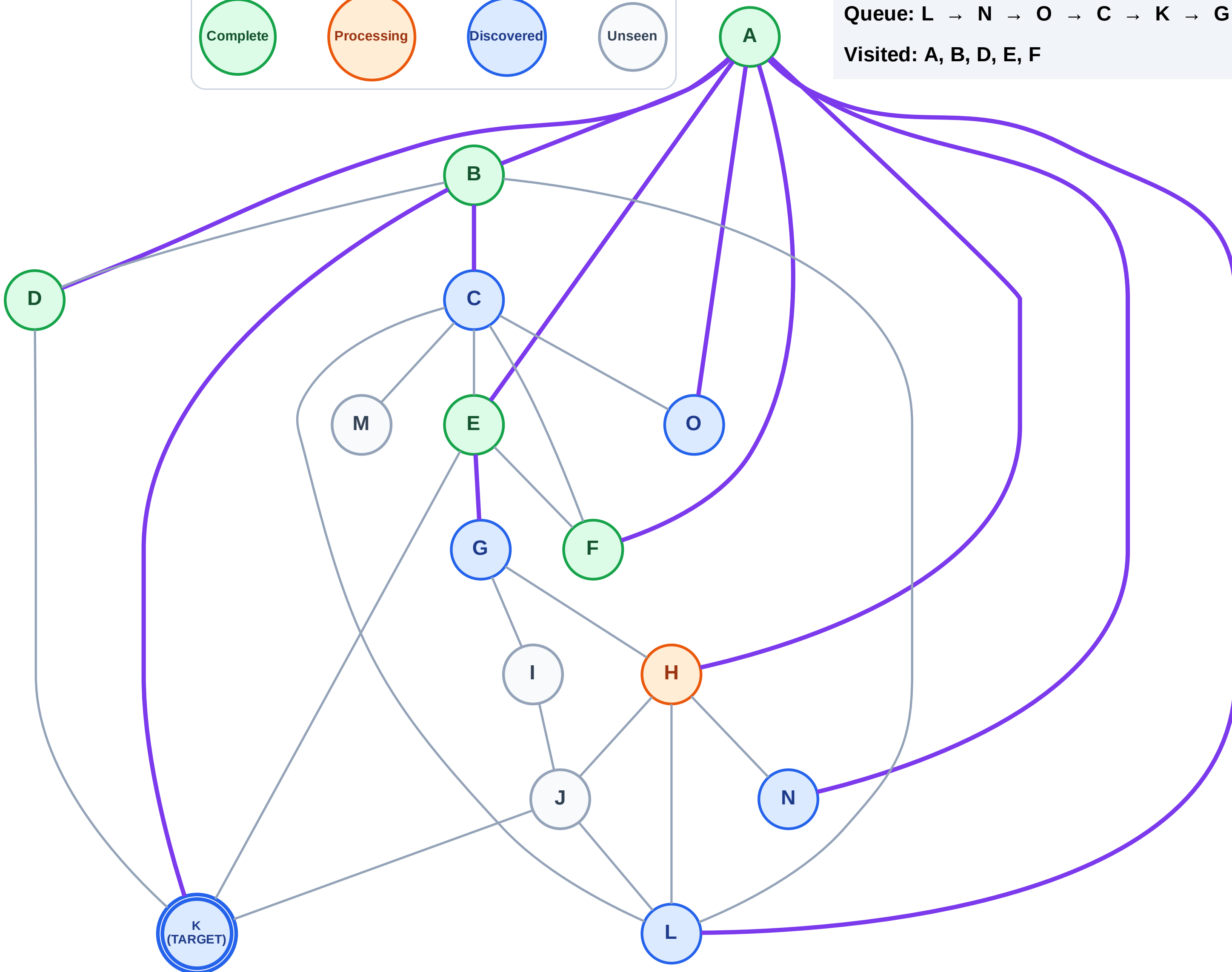
Step 12: Process node H

Legend



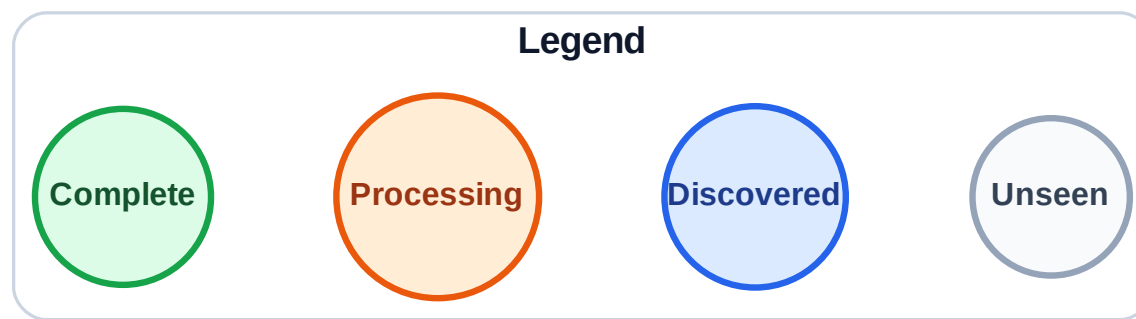
Queue: L → N → O → C → K → G

Visited: A, B, D, E, F



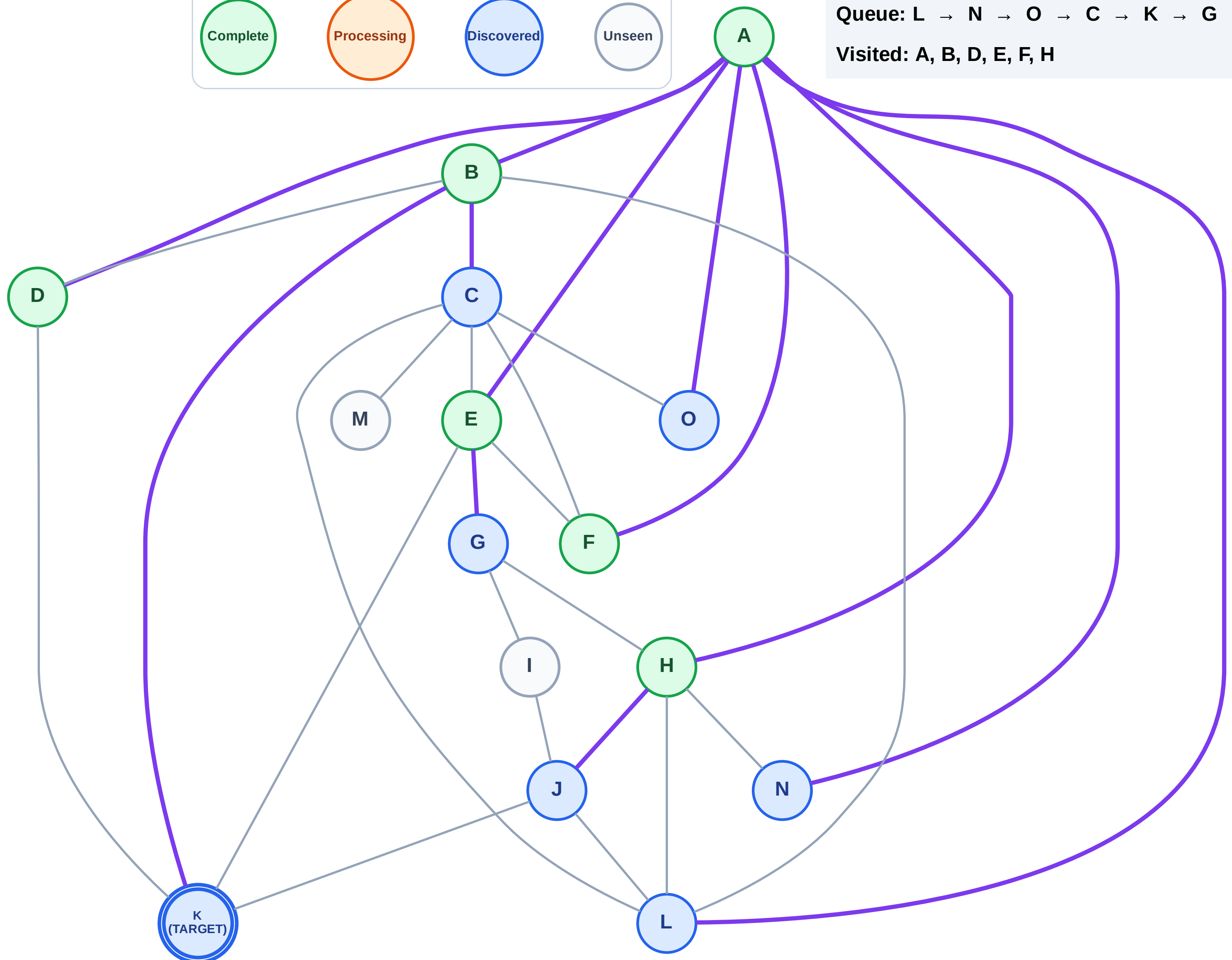
BFS Traversal

Step 13: Discover J from H



Queue: L → N → O → C → K → G → J

Visited: A, B, D, E, F, H



BFS Traversal

Step 14: Process node L

Legend

Complete

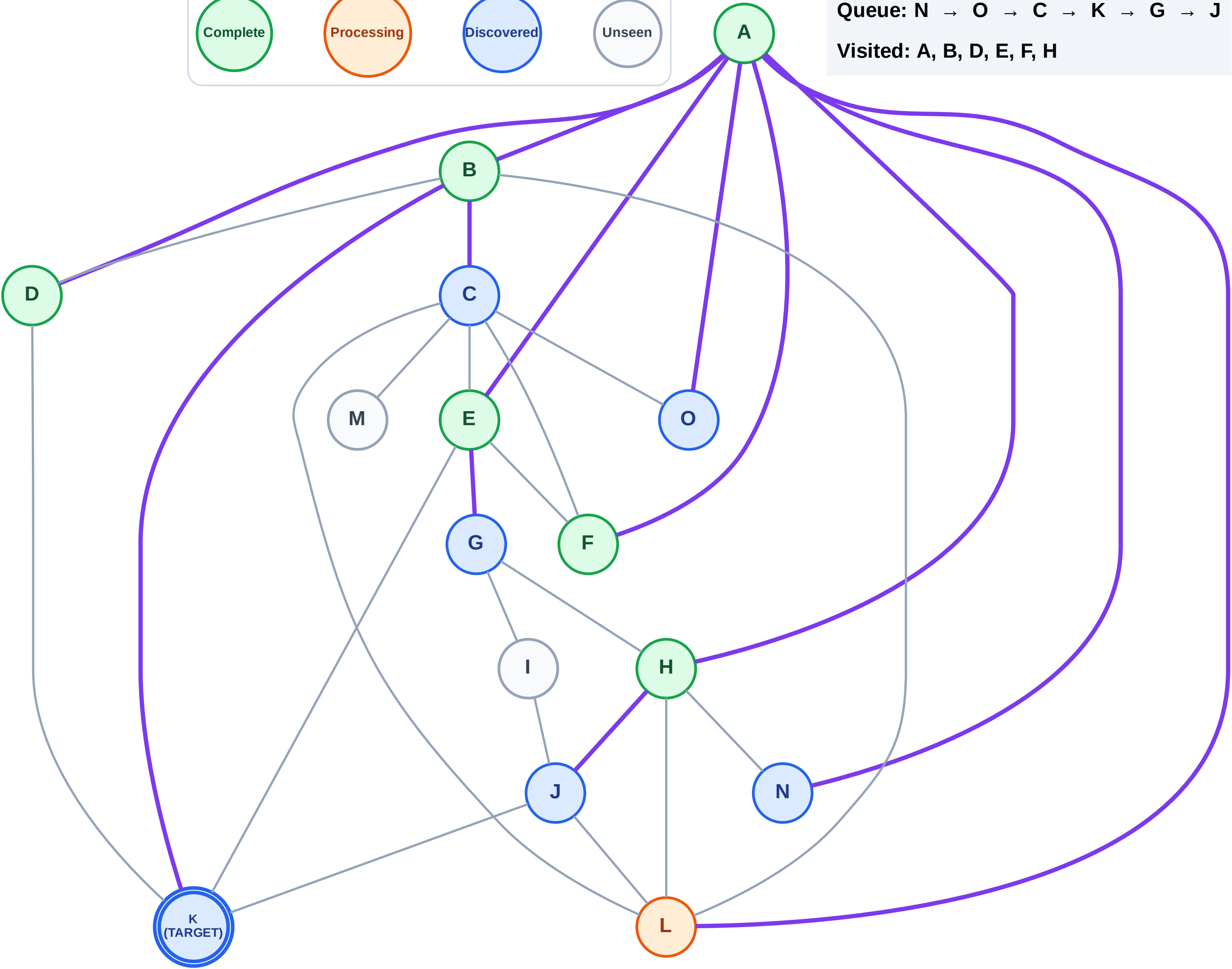
Processing

Discovered

Unseen

Queue: N → O → C → K → G → J

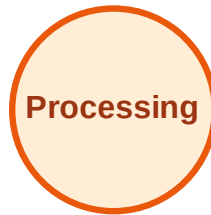
Visited: A, B, D, E, F, H



BFS Traversal

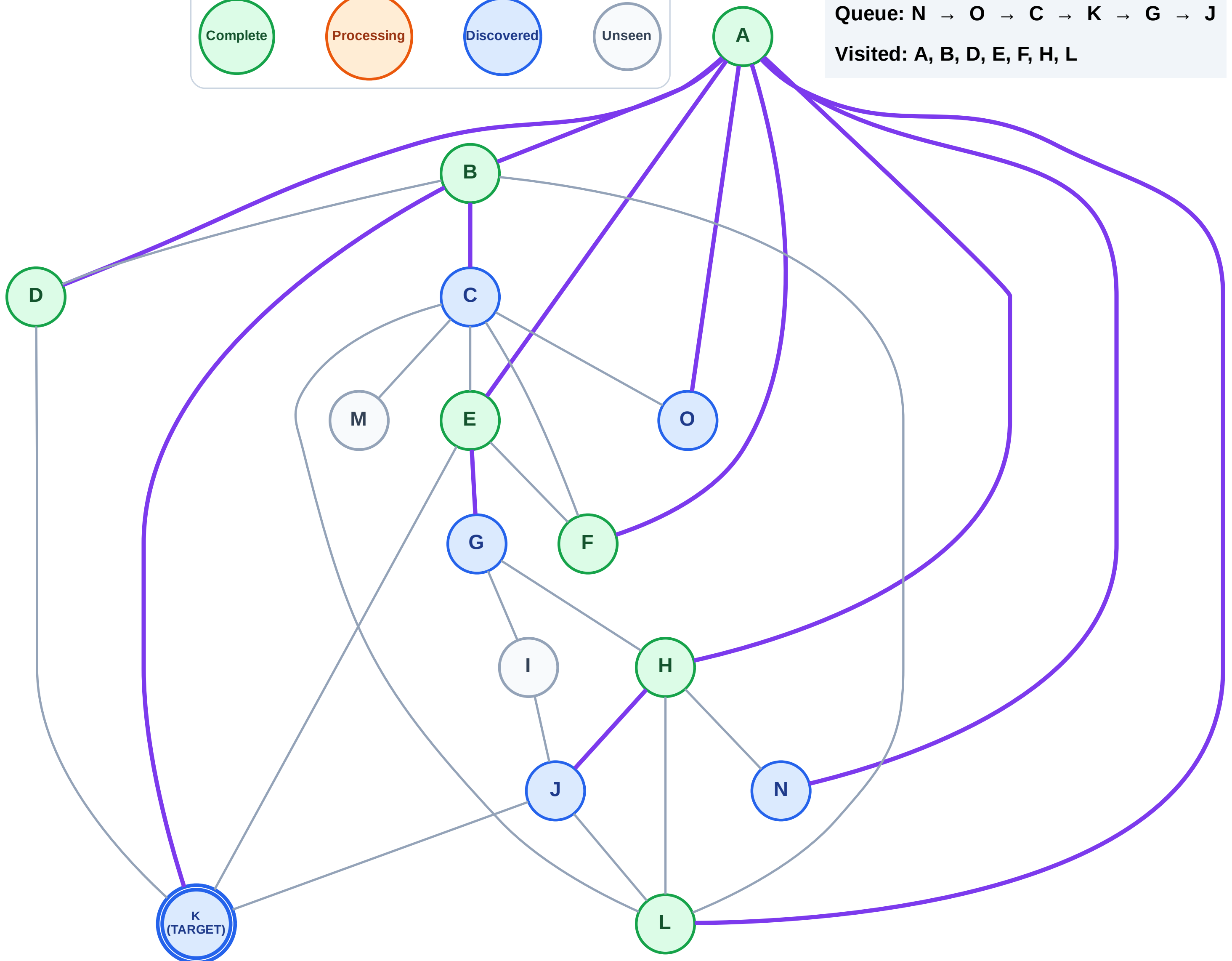
Step 15: No new neighbors from L

Legend



Queue: N → O → C → K → G → J

Visited: A, B, D, E, F, H, L



BFS Traversal

Step 16: Process node N

Legend



Complete



Processing



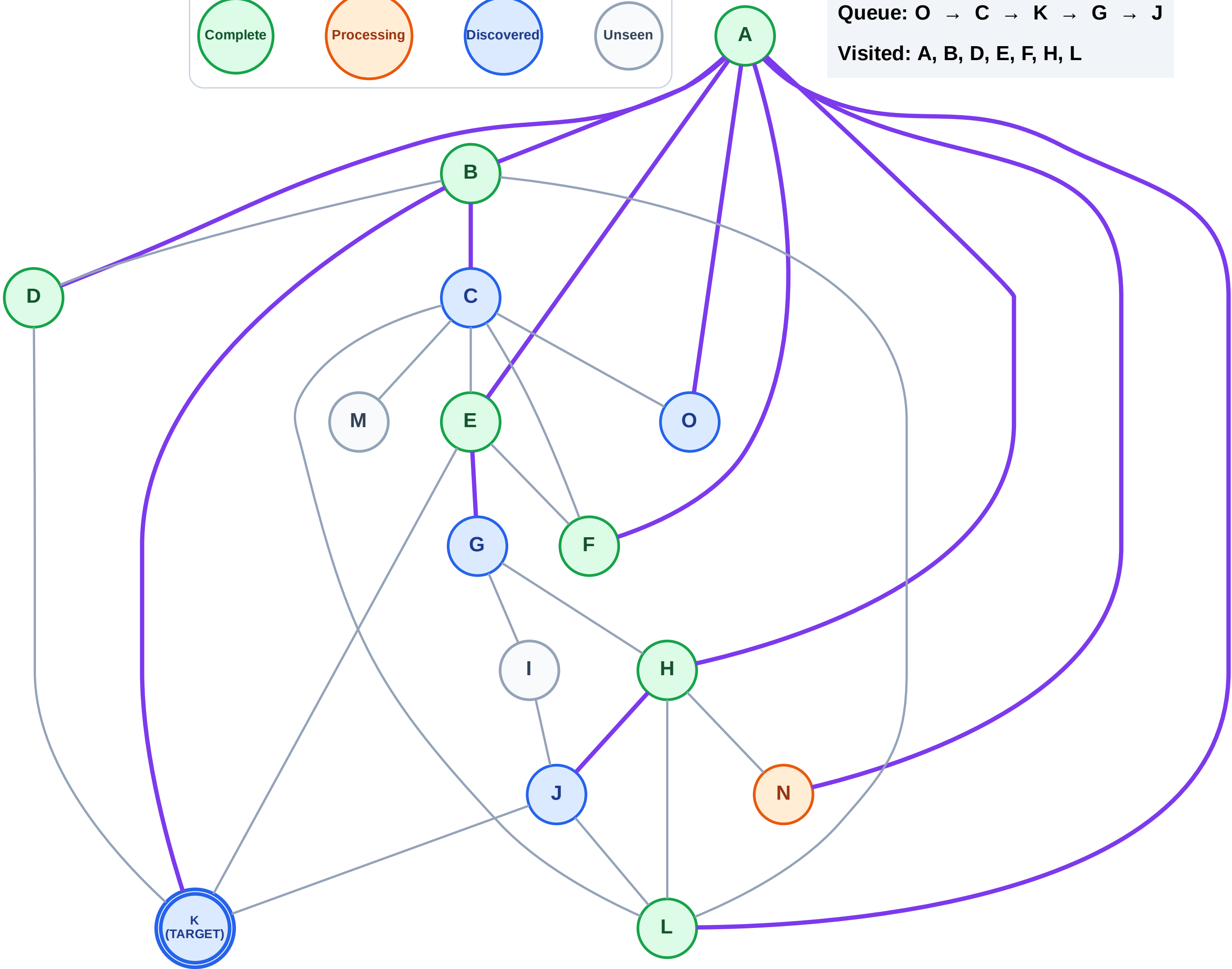
Discovered



Unseen

Queue: O → C → K → G → J

Visited: A, B, D, E, F, H, L



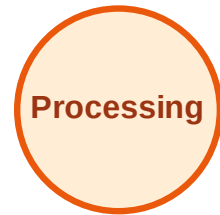
BFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



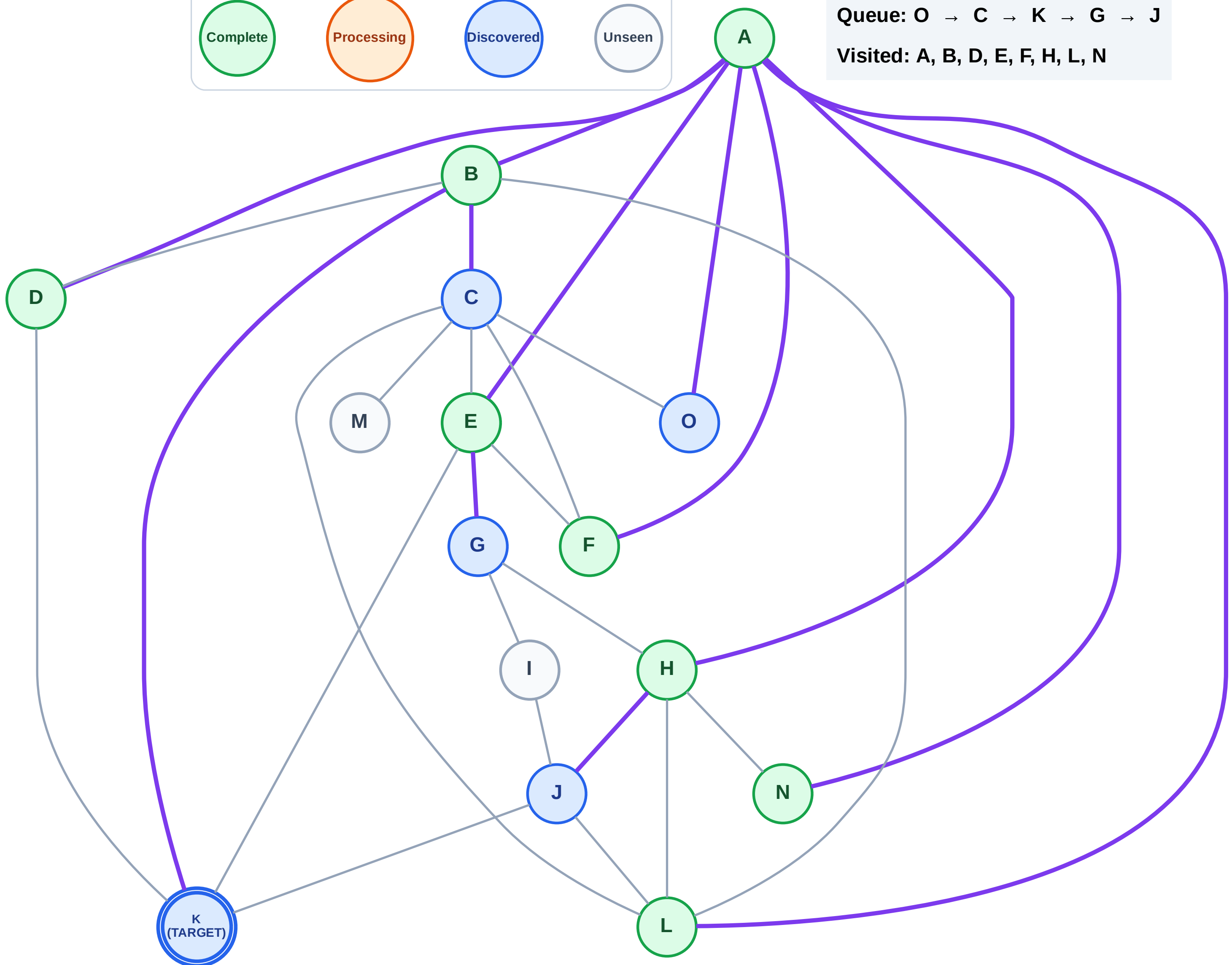
Discovered



Unseen

Queue: O → C → K → G → J

Visited: A, B, D, E, F, H, L, N



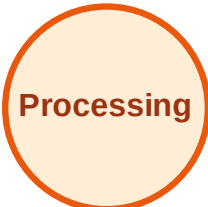
BFS Traversal

Step 18: Process node O

Legend



Complete



Processing



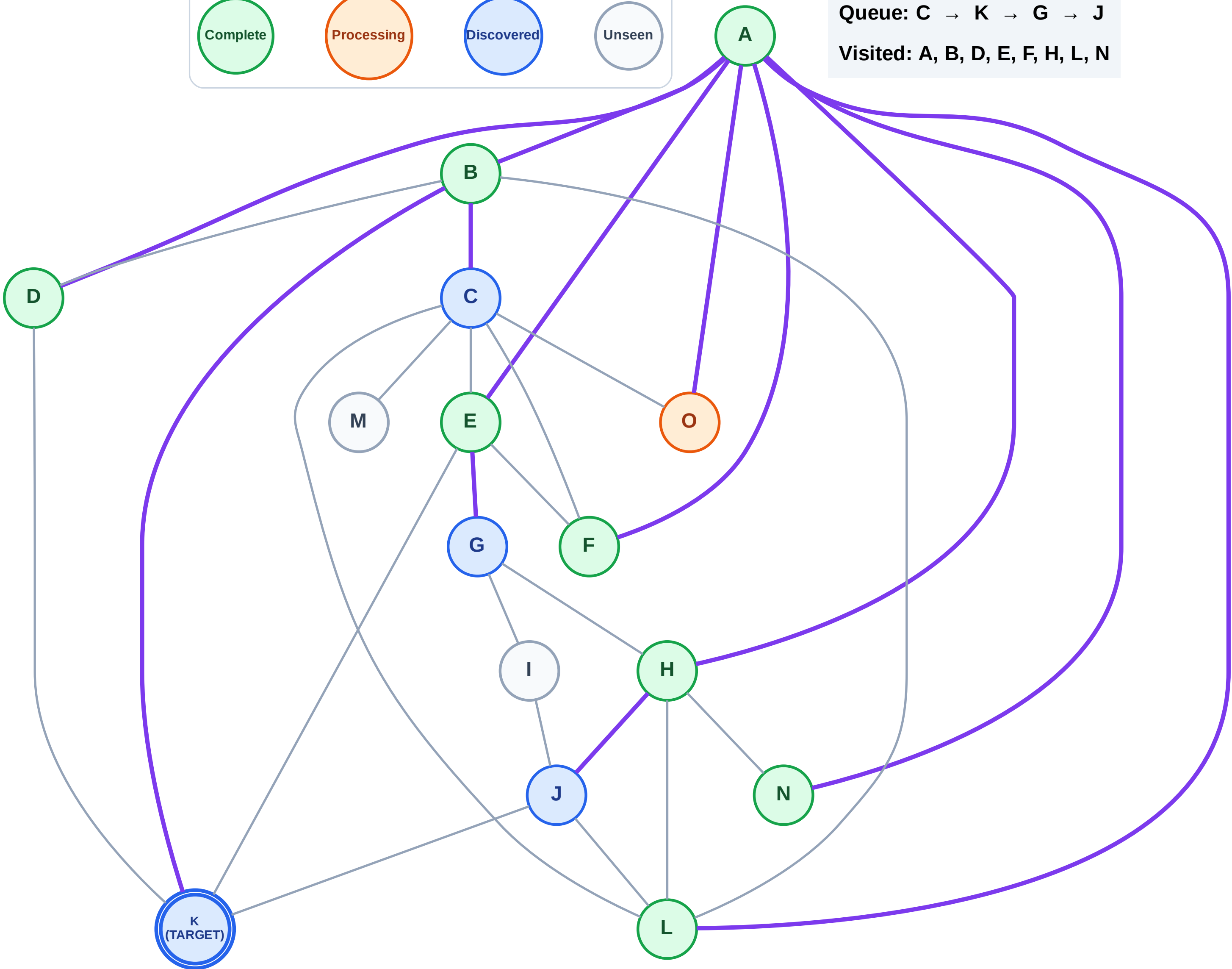
Discovered



Unseen

Queue: C → K → G → J

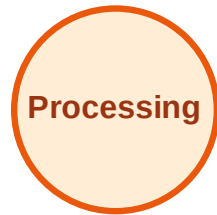
Visited: A, B, D, E, F, H, L, N



BFS Traversal

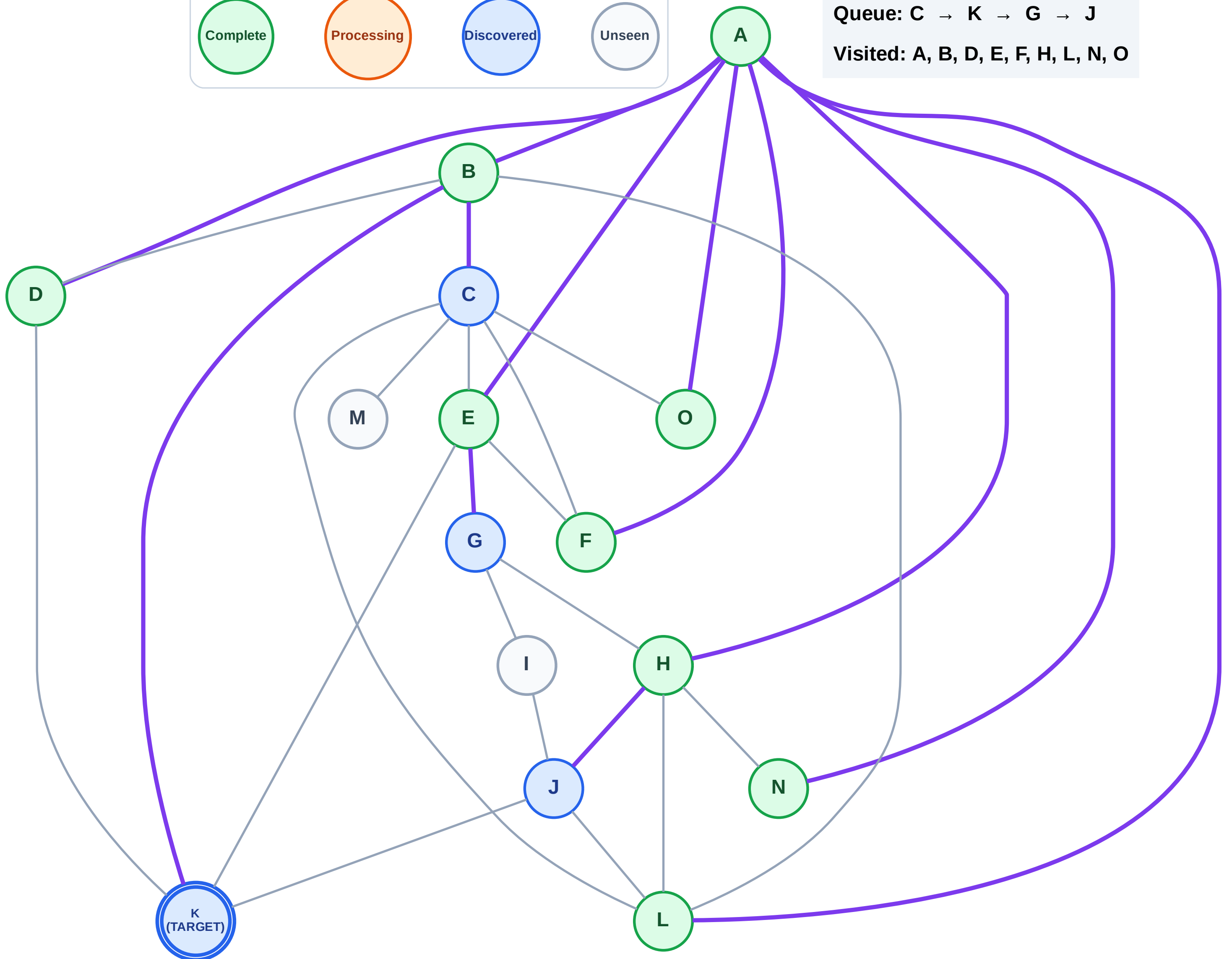
Step 19: No new neighbors from O

Legend



Queue: C → K → G → J

Visited: A, B, D, E, F, H, L, N, O



BFS Traversal

Step 20: Process node C

Legend



Complete



Processing



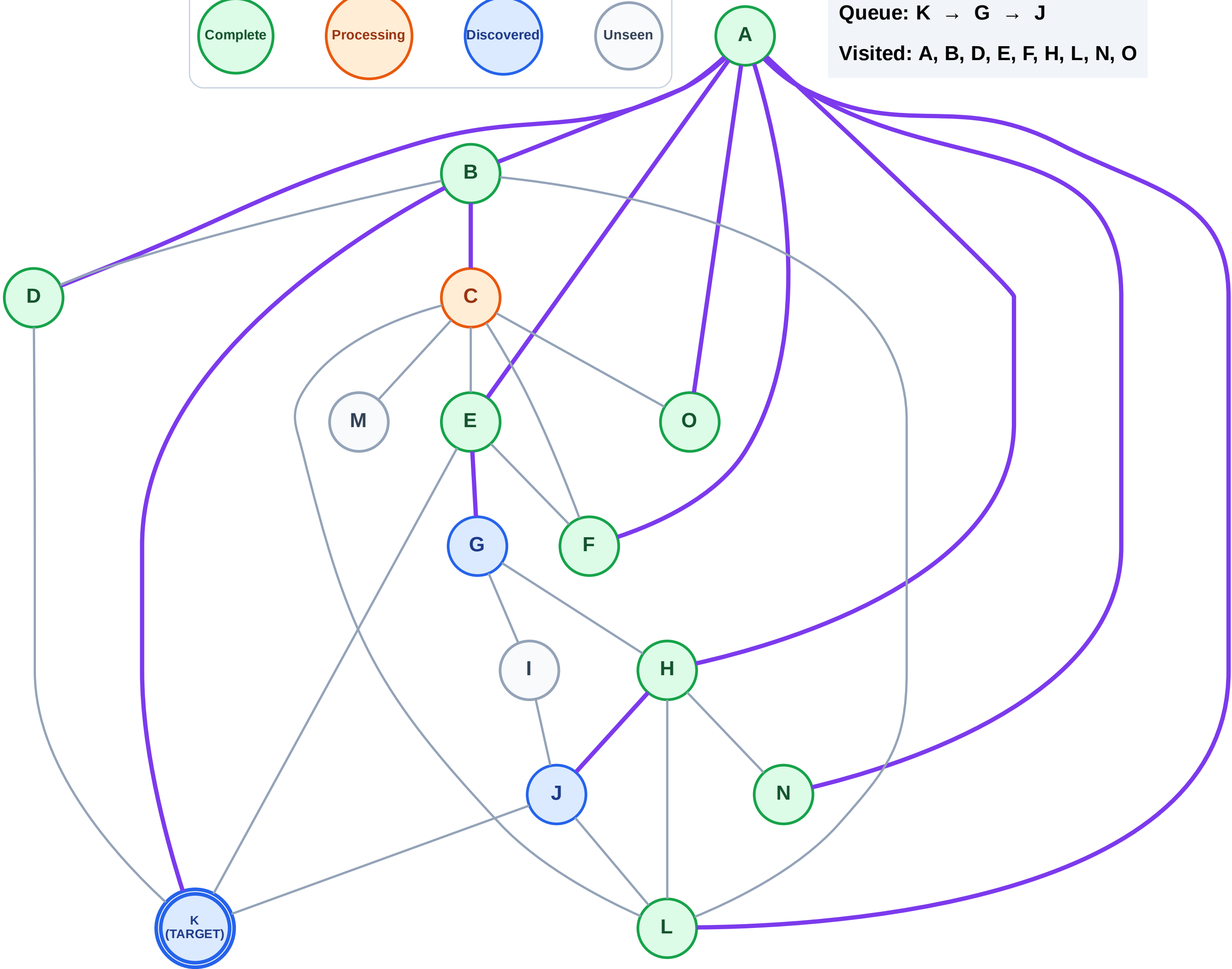
Discovered



Unseen

Queue: K → G → J

Visited: A, B, D, E, F, H, L, N, O



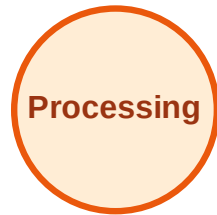
BFS Traversal

Step 21: Discover M from C

Legend



Complete



Processing



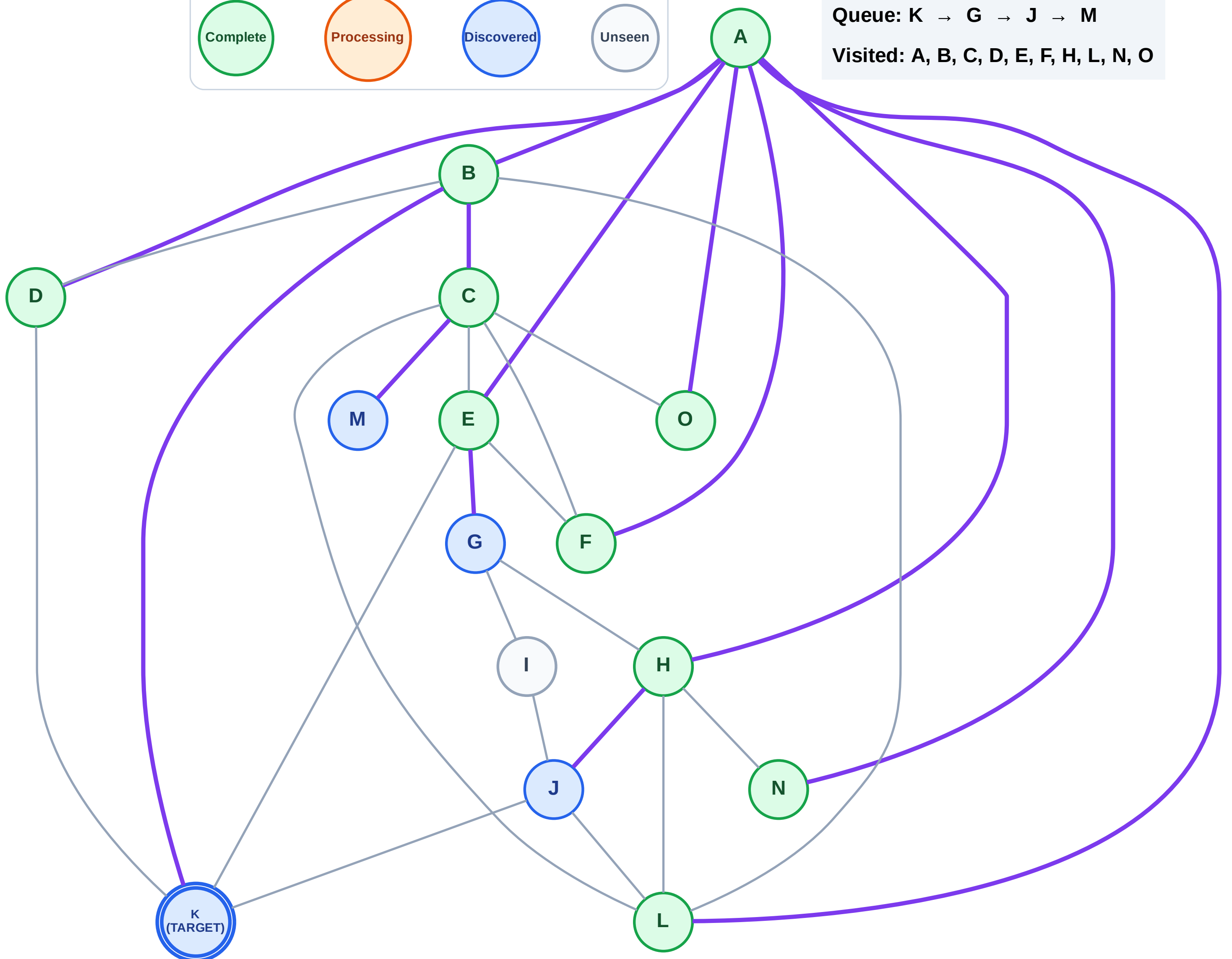
Discovered



Unseen

Queue: K → G → J → M

Visited: A, B, C, D, E, F, H, L, N, O



BFS Traversal

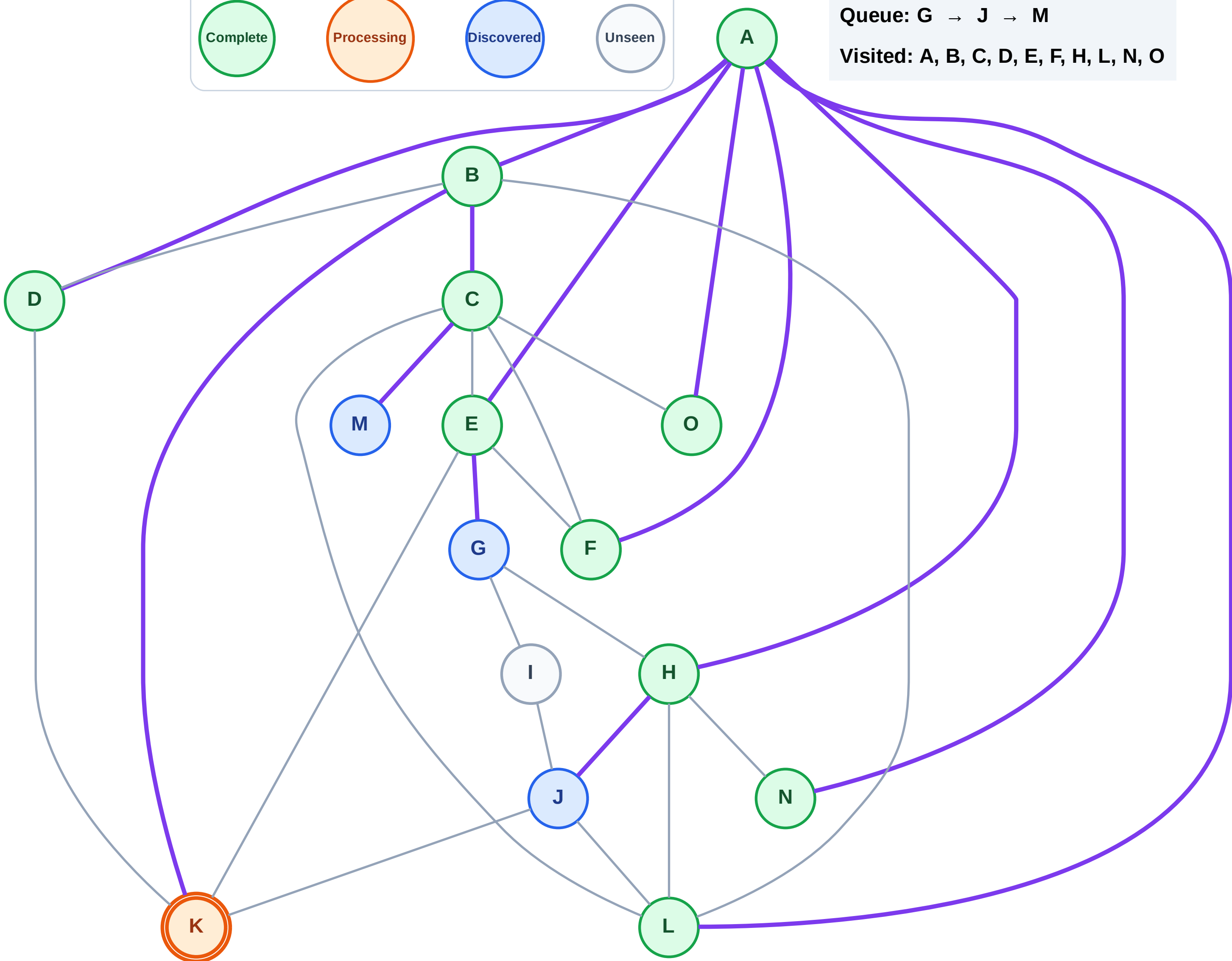
Step 22: Process node K

Legend



Queue: G → J → M

Visited: A, B, C, D, E, F, H, L, N, O



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

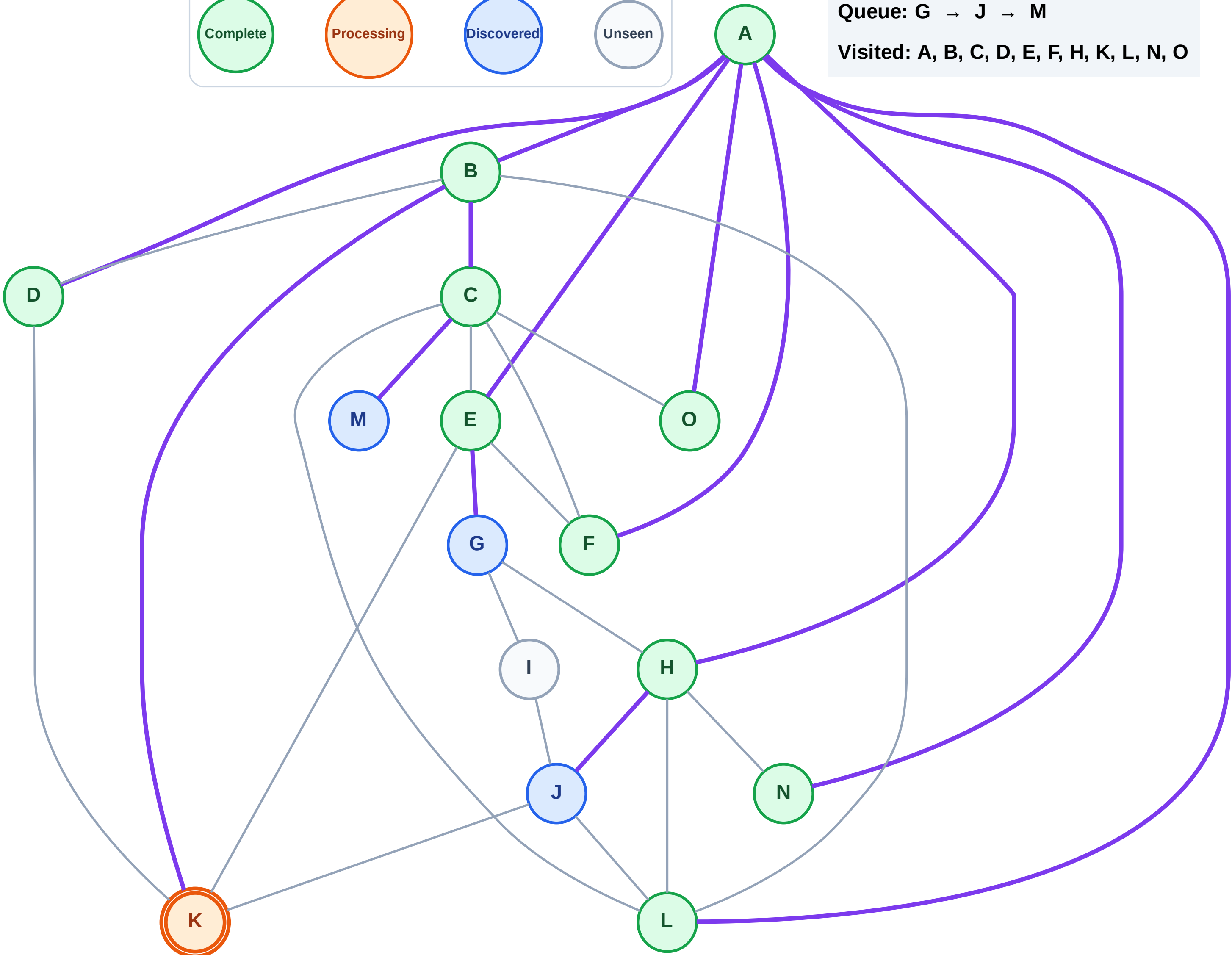
Processing

Discovered

Unseen

Queue: G → J → M

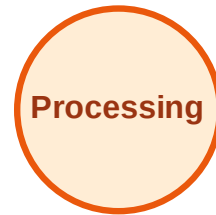
Visited: A, B, C, D, E, F, H, K, L, N, O



DFS Traversal

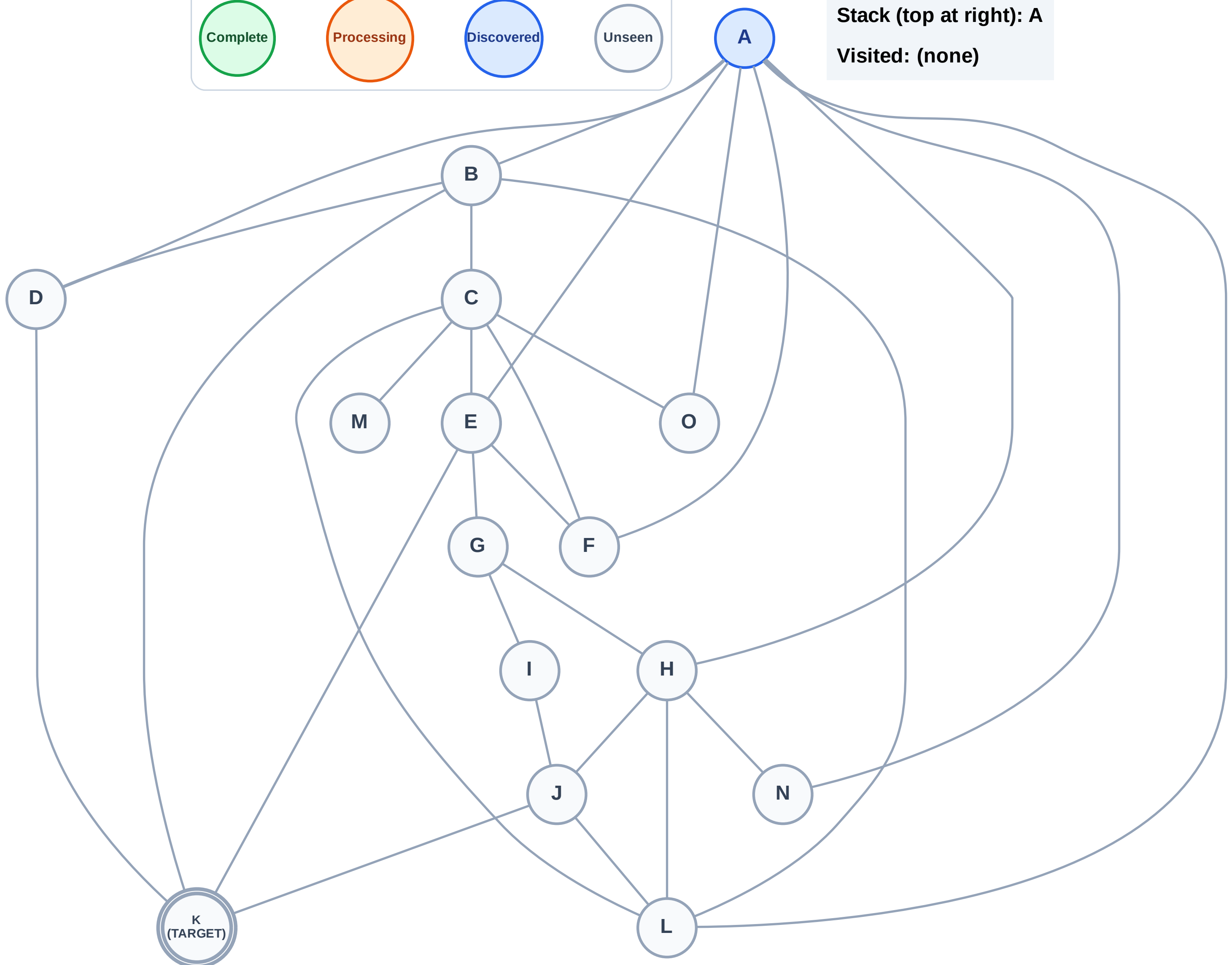
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

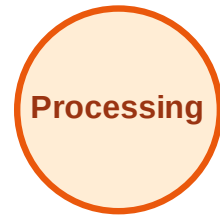
Visited: (none)



DFS Traversal

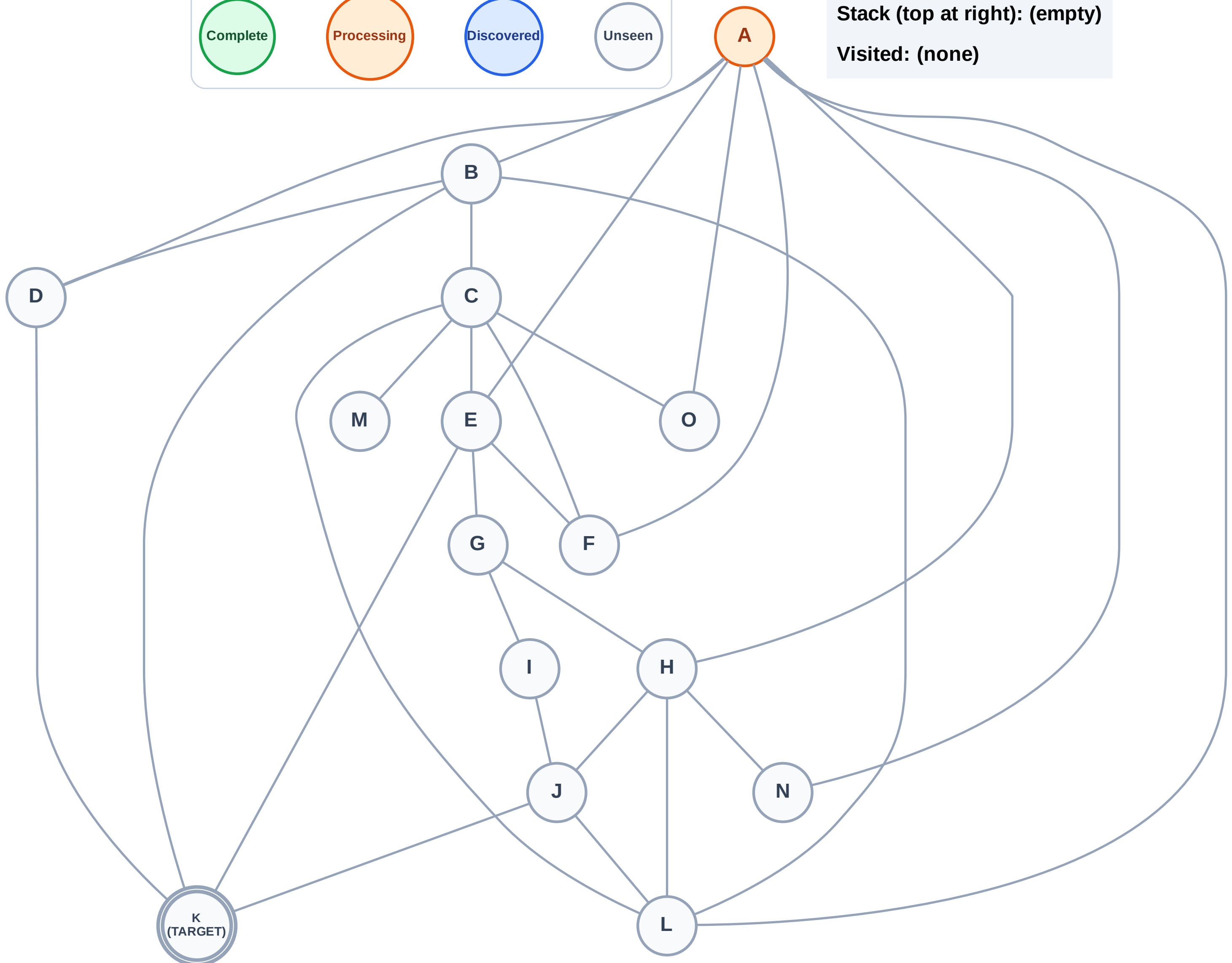
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover O, N, L, H, F, E, D, B from A

Legend

Complete

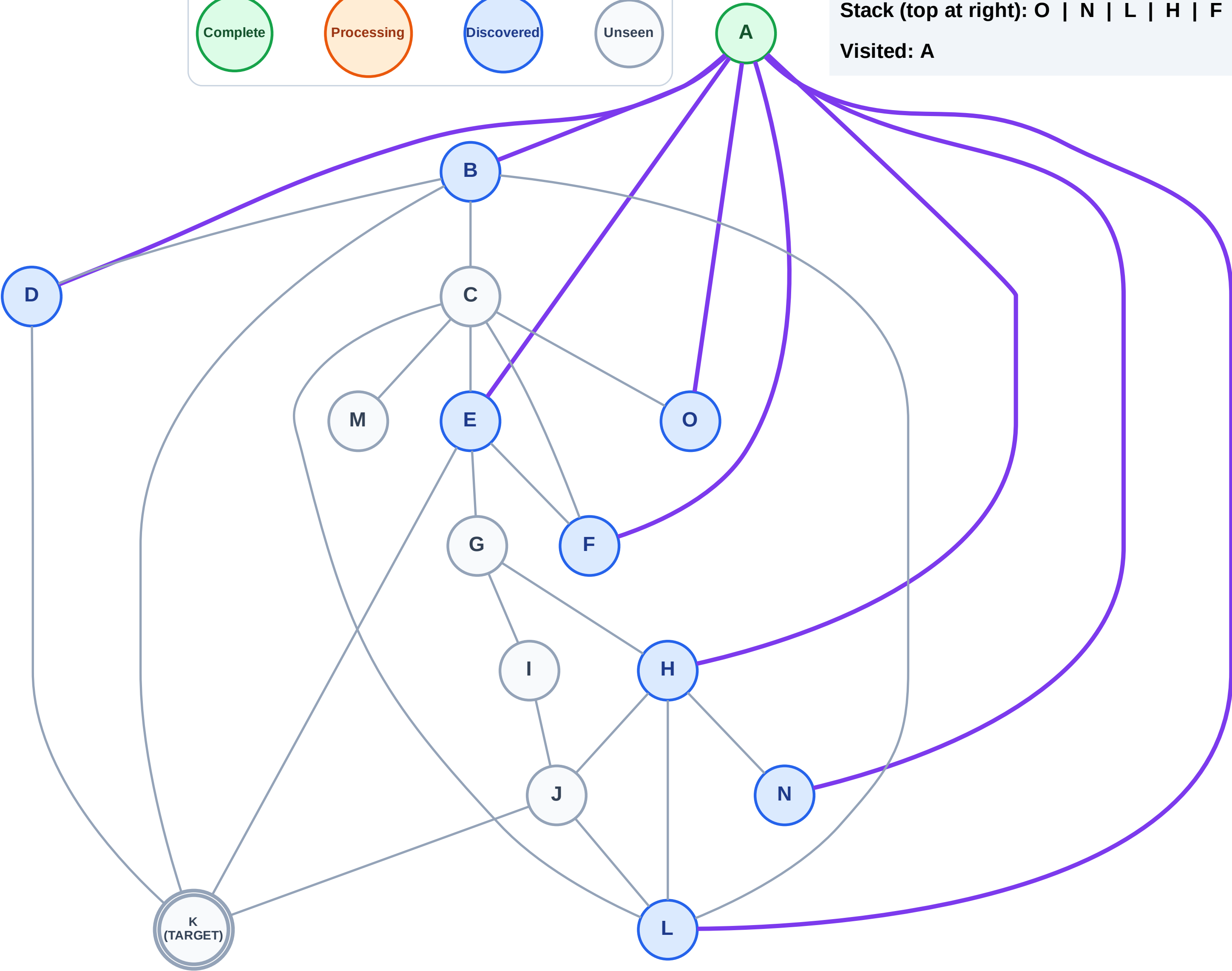
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

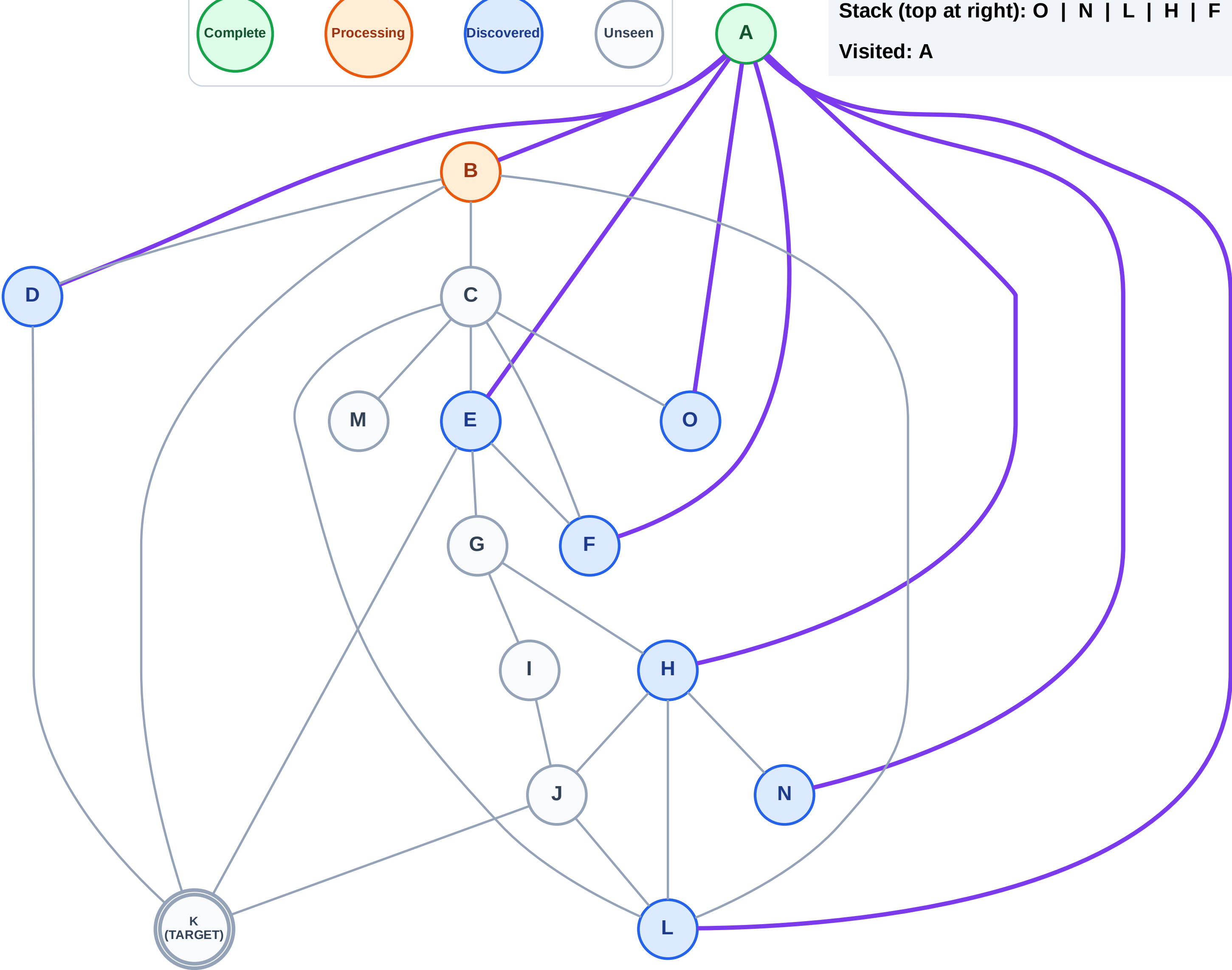
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D

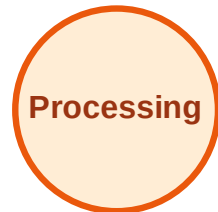
Visited: A



DFS Traversal

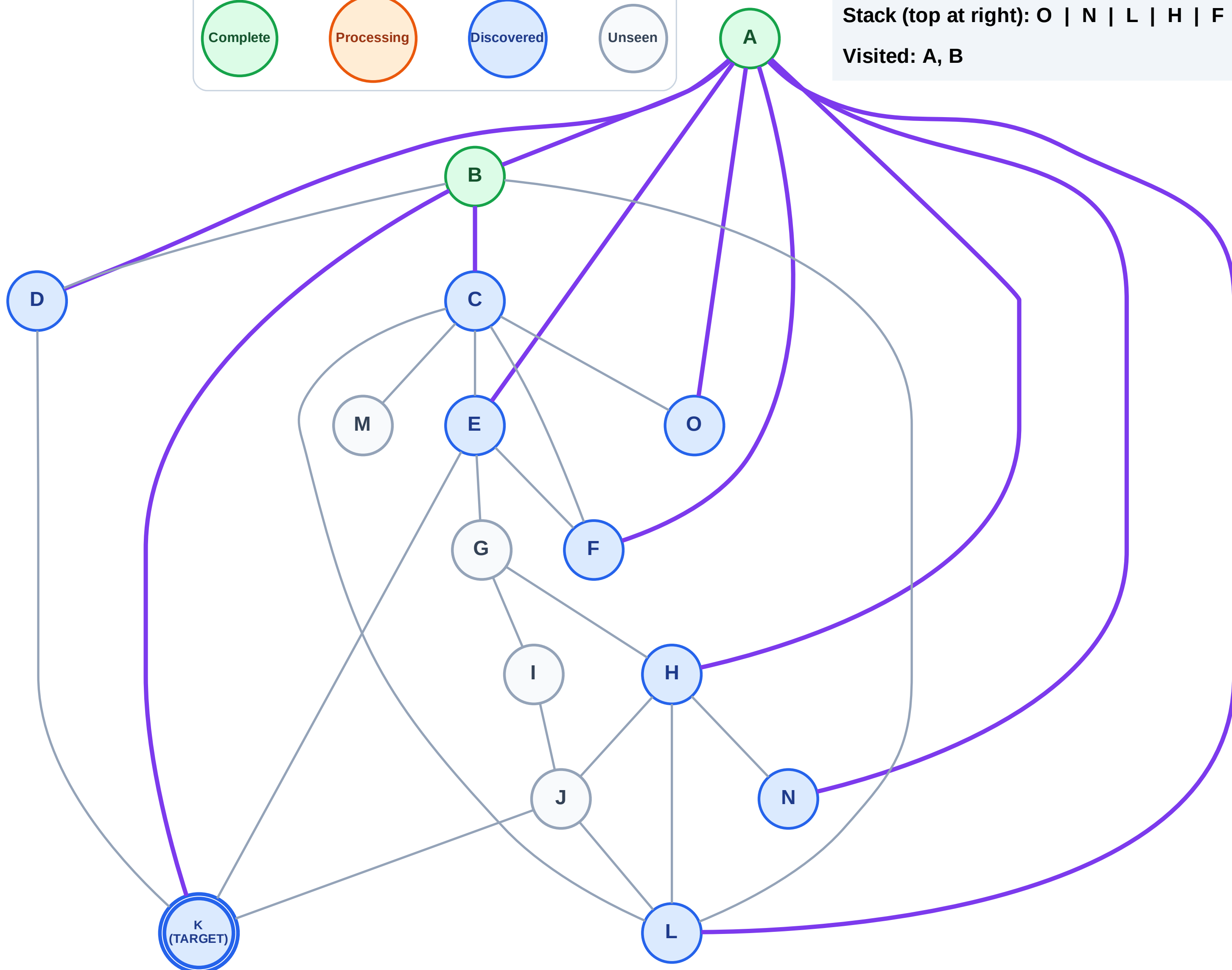
Step 5: Discover K, C from B

Legend



Stack (top at right): O | N | L | H | F | E | D | K | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

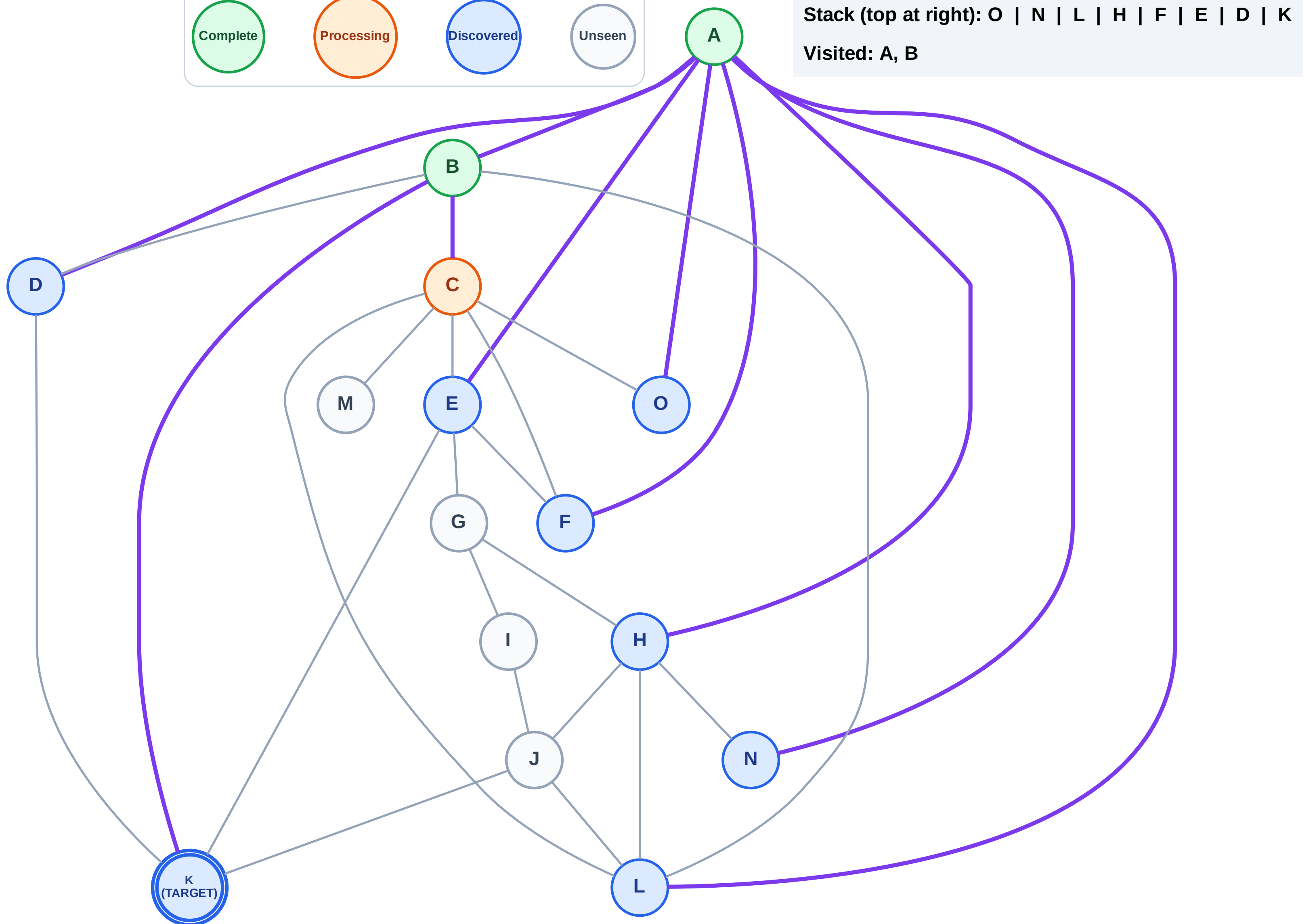
Processing

Discovered

Unseen

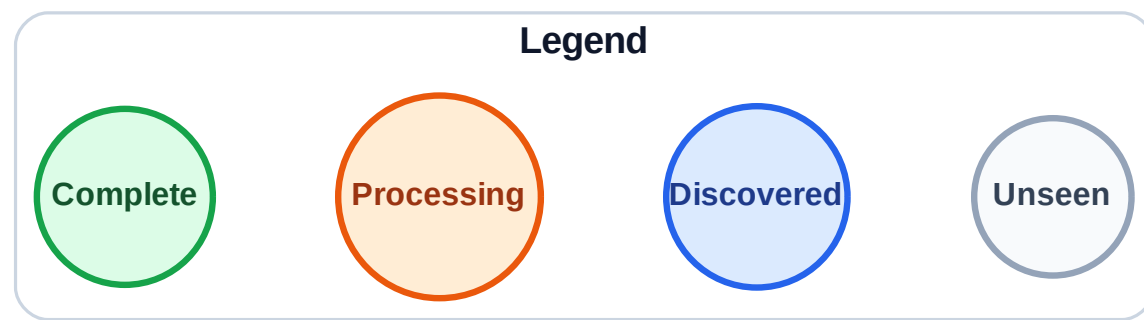
Stack (top at right): O | N | L | H | F | E | D | K

Visited: A, B



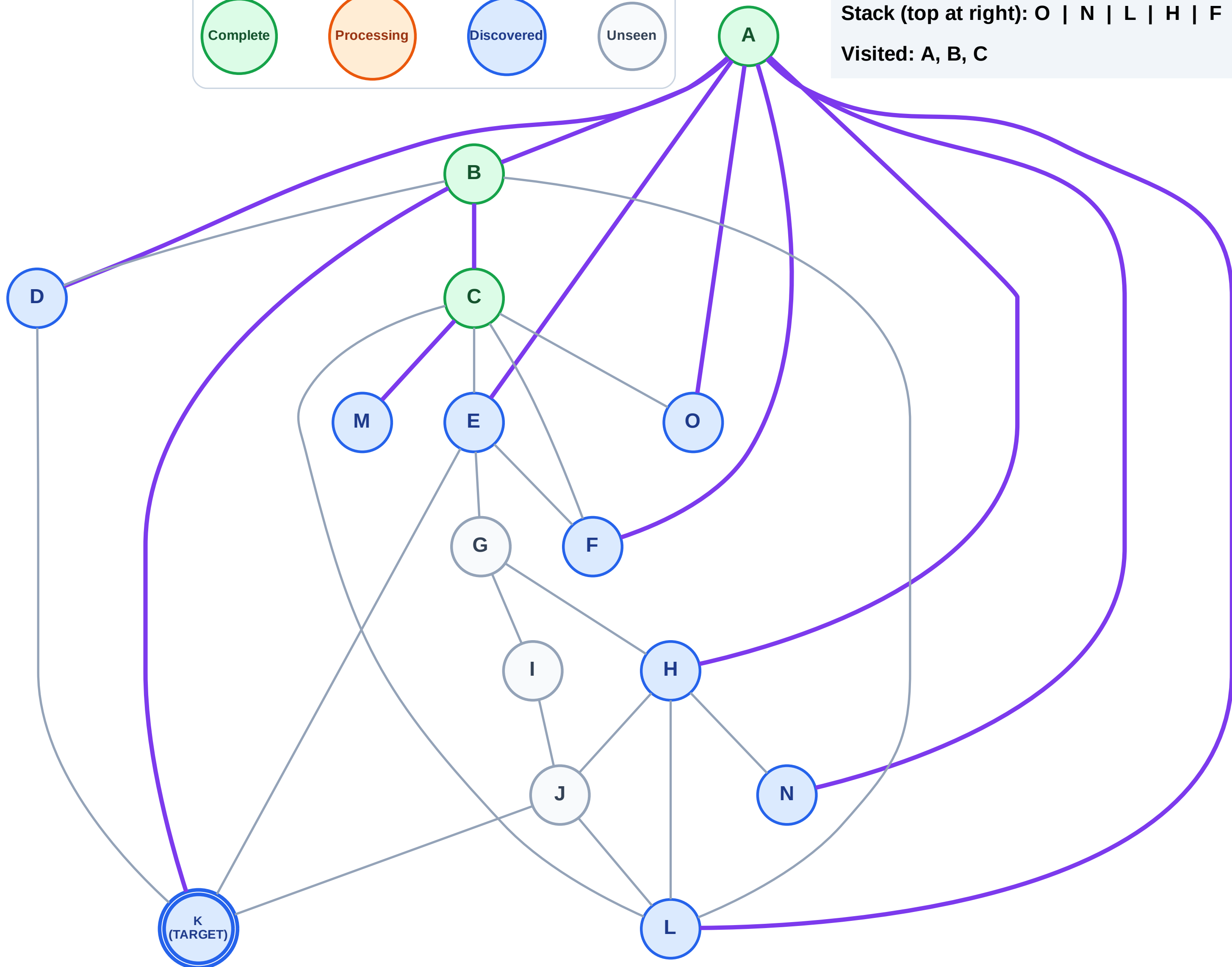
DFS Traversal

Step 7: Discover M from C



Stack (top at right): O | N | L | H | F | E | D | K | M

Visited: A, B, C



DFS Traversal

Step 8: Process node M

Legend

Complete

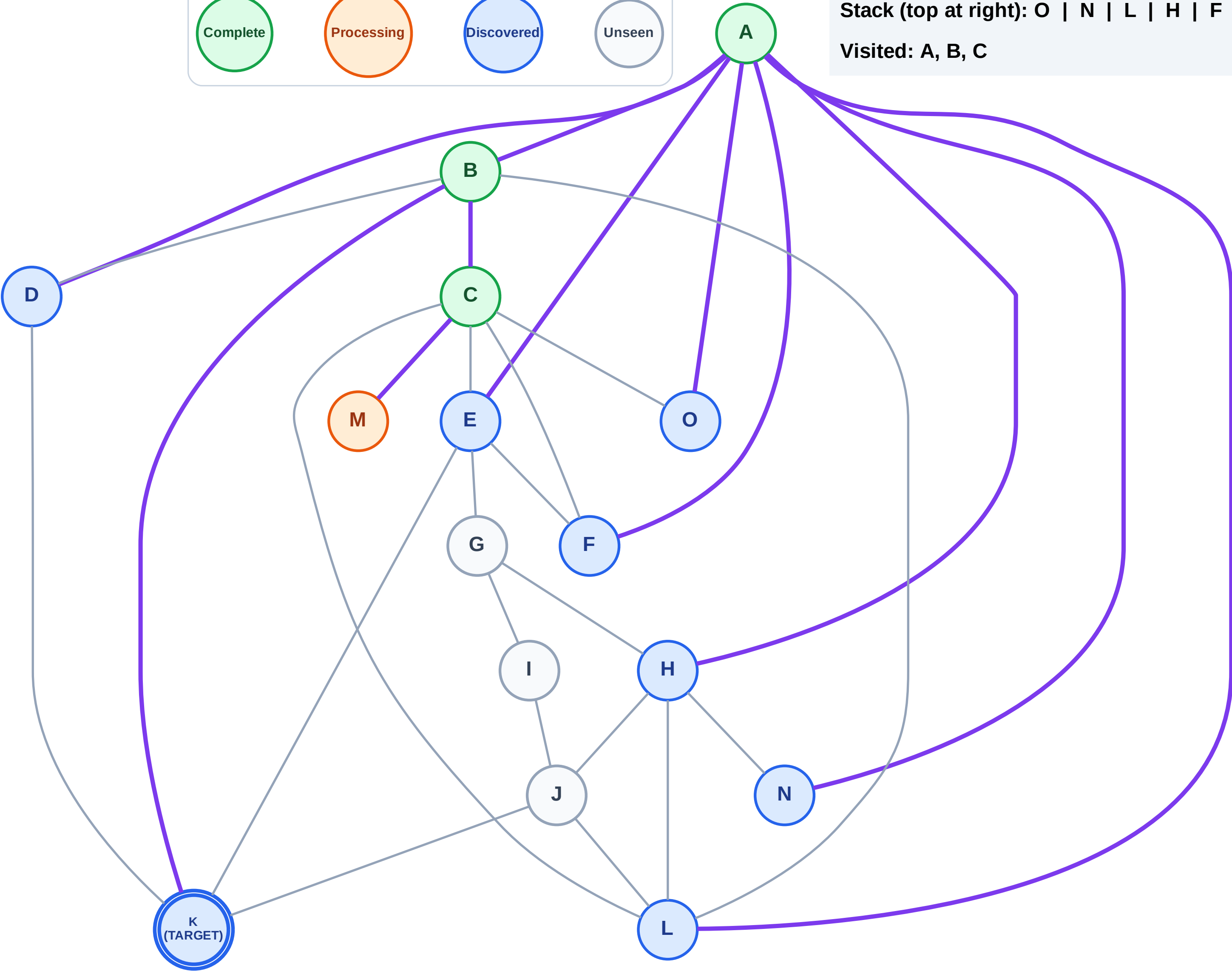
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D | K

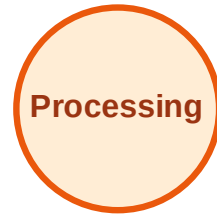
Visited: A, B, C



DFS Traversal

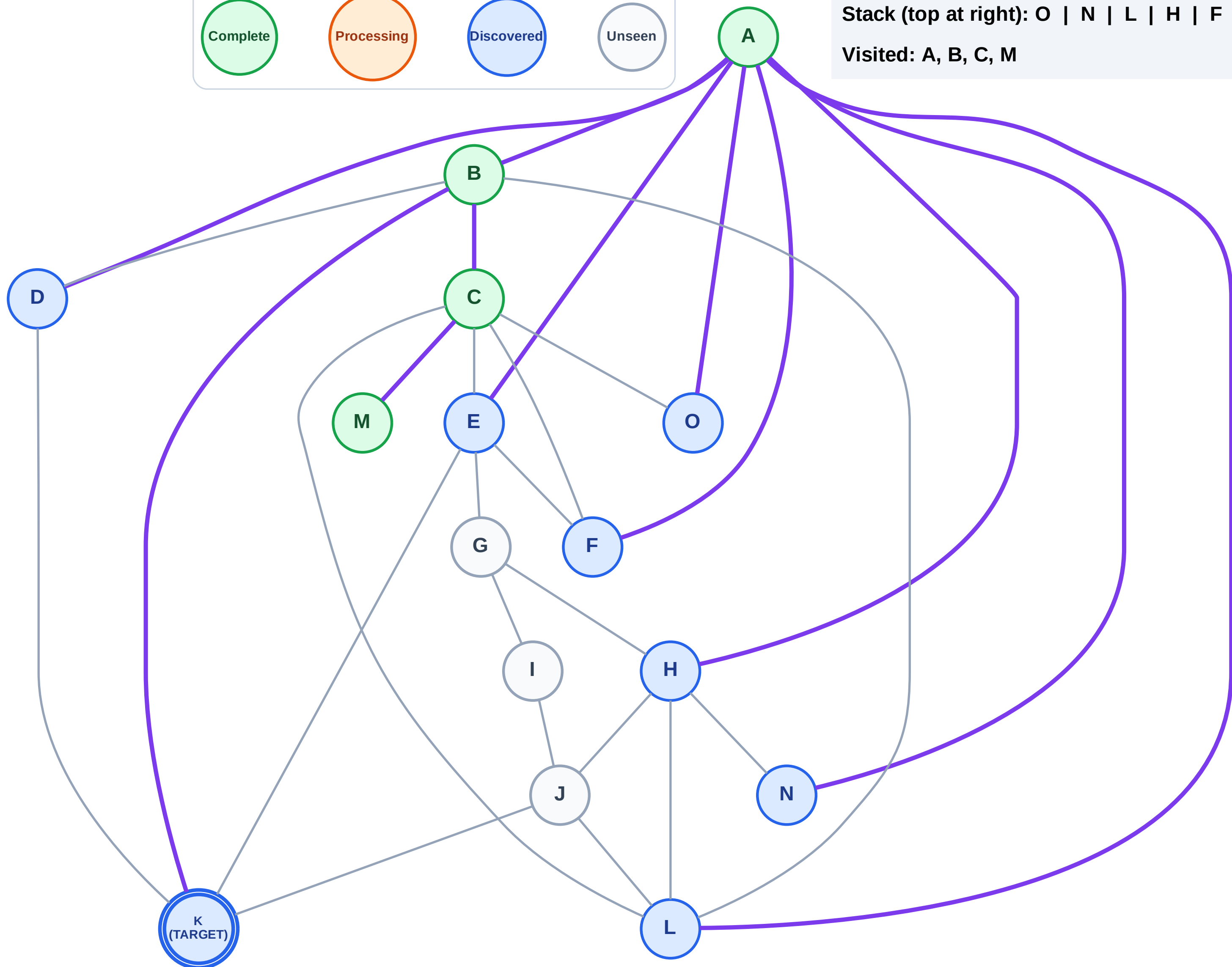
Step 9: No new neighbors from M

Legend



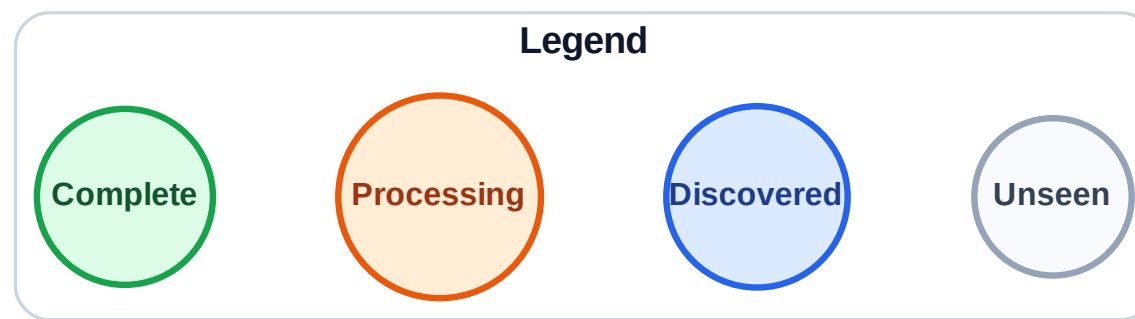
Stack (top at right): O | N | L | H | F | E | D | K

Visited: A, B, C, M



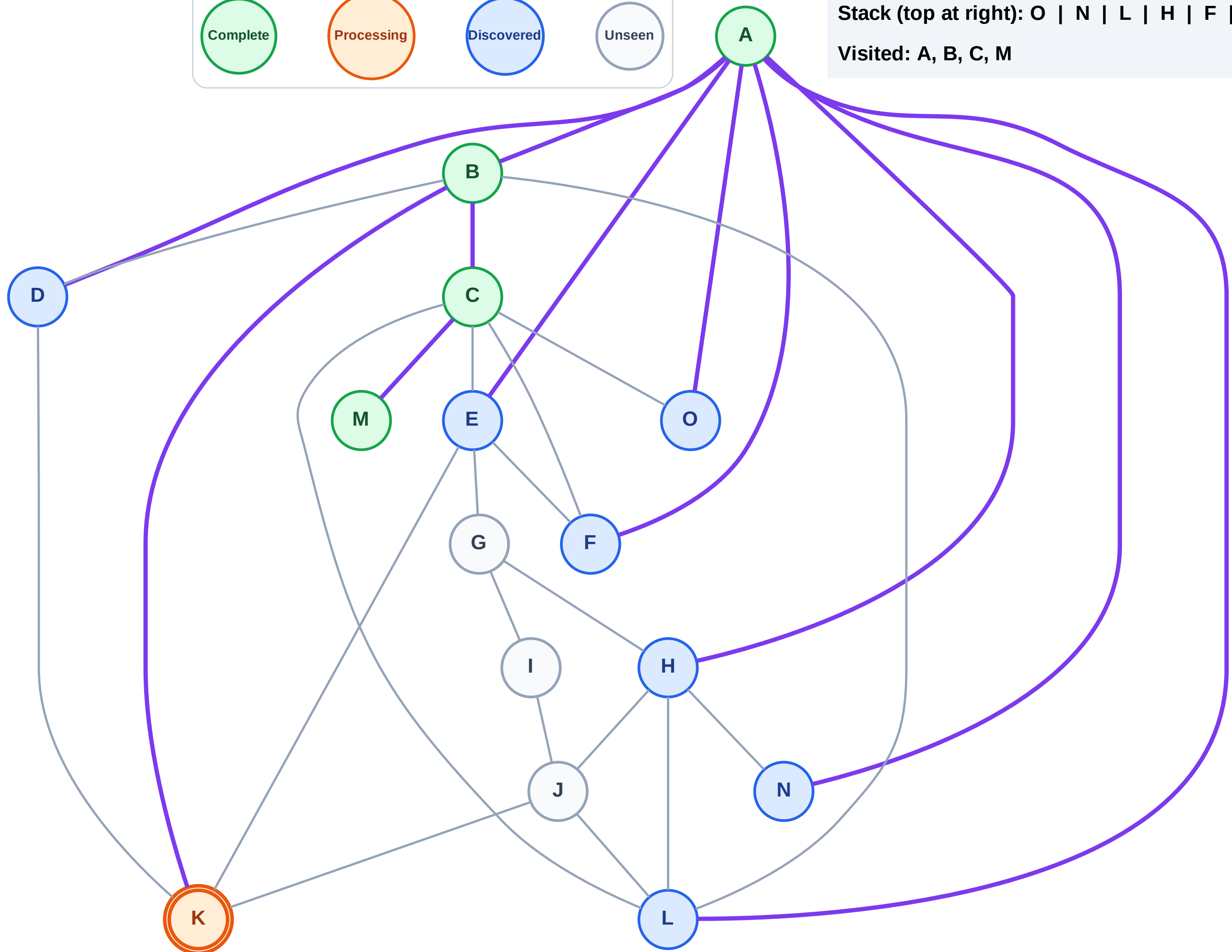
DFS Traversal

Step 10: Process node K



Stack (top at right): O | N | L | H | F | E | D

Visited: A, B, C, M



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D

Visited: A, B, C, K, M

